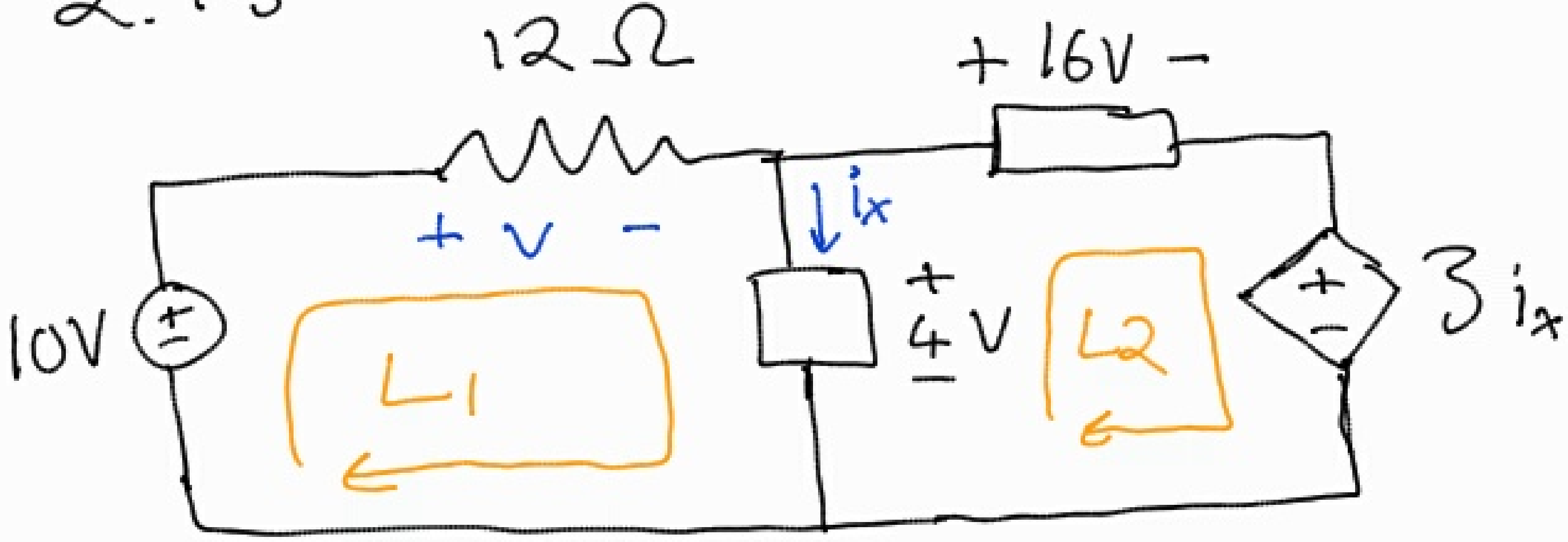
Tutorial 2

2.15



KVL : $\sum v_i = 0$

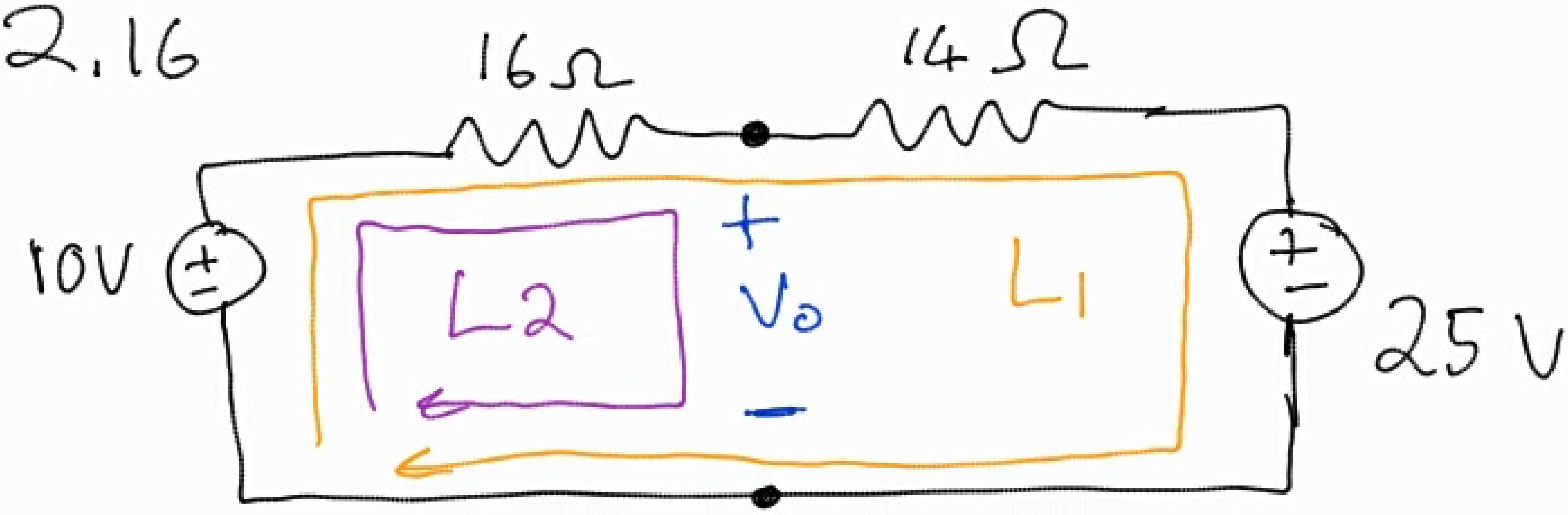
Loop 1:

$$-10 + v + 4 = 0$$
$$v = 10 - 4$$
$$v = 6 \text{ V}$$

Loop 2:

$$-4 + 16 + 3i_x = 0$$
$$3i_x = -12$$
$$i_x = -4 \text{ A}$$

2.16



Loop 1: Use KVL and Ohm's Law to find the current.

$$-10 + 16I + 14I + 25 = 0$$

$$30I = -15$$

$$I = -0{,}5 \text{ A}$$

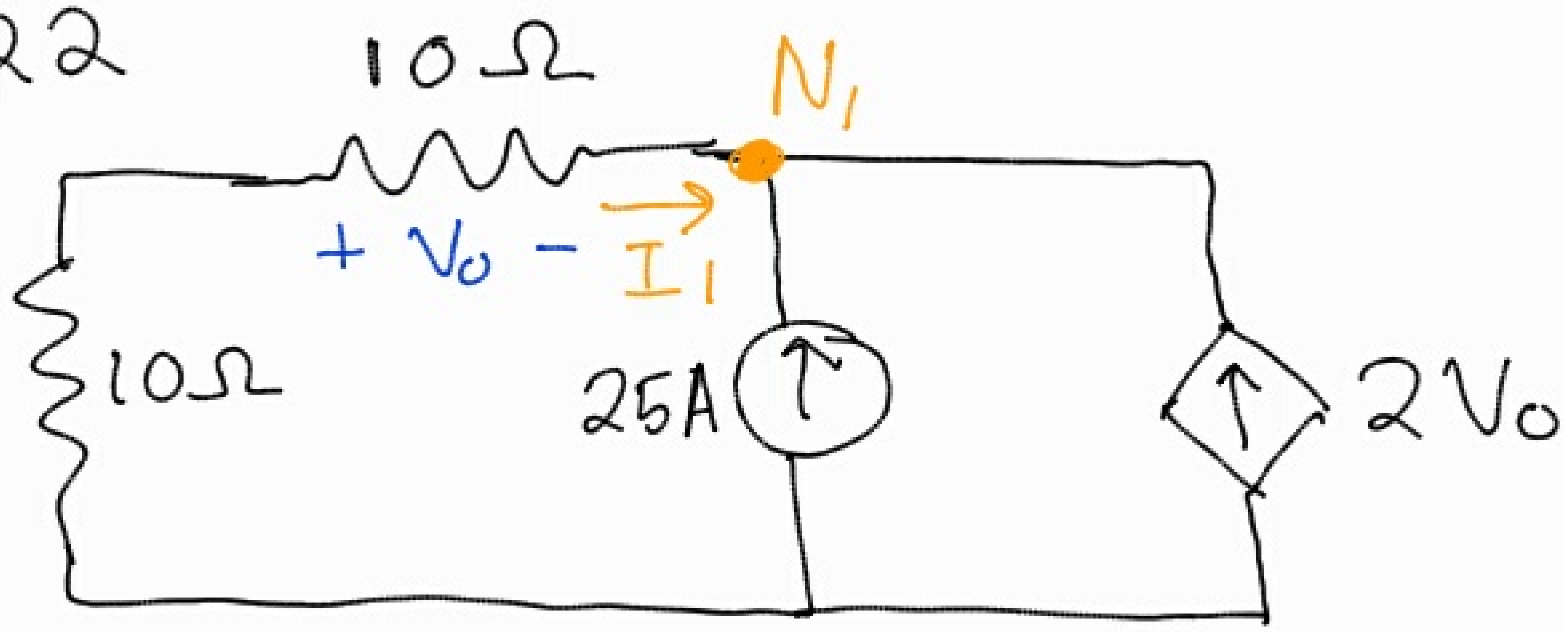
Loop 2: Use KVL & I

$$-10 + 16I + V_o = 0$$

$$-10 - 8 + V_o = 0$$

$$V_o = 18 \text{ V}$$

2.22



Apply KCL at node 1

$$\sum I_n = 0$$

$$25 + 2V_0 + I_1 = 0$$

$$I_1 = -25 - 2V_0$$

Use ohm's Law at V_0

$$V = I \cdot R$$

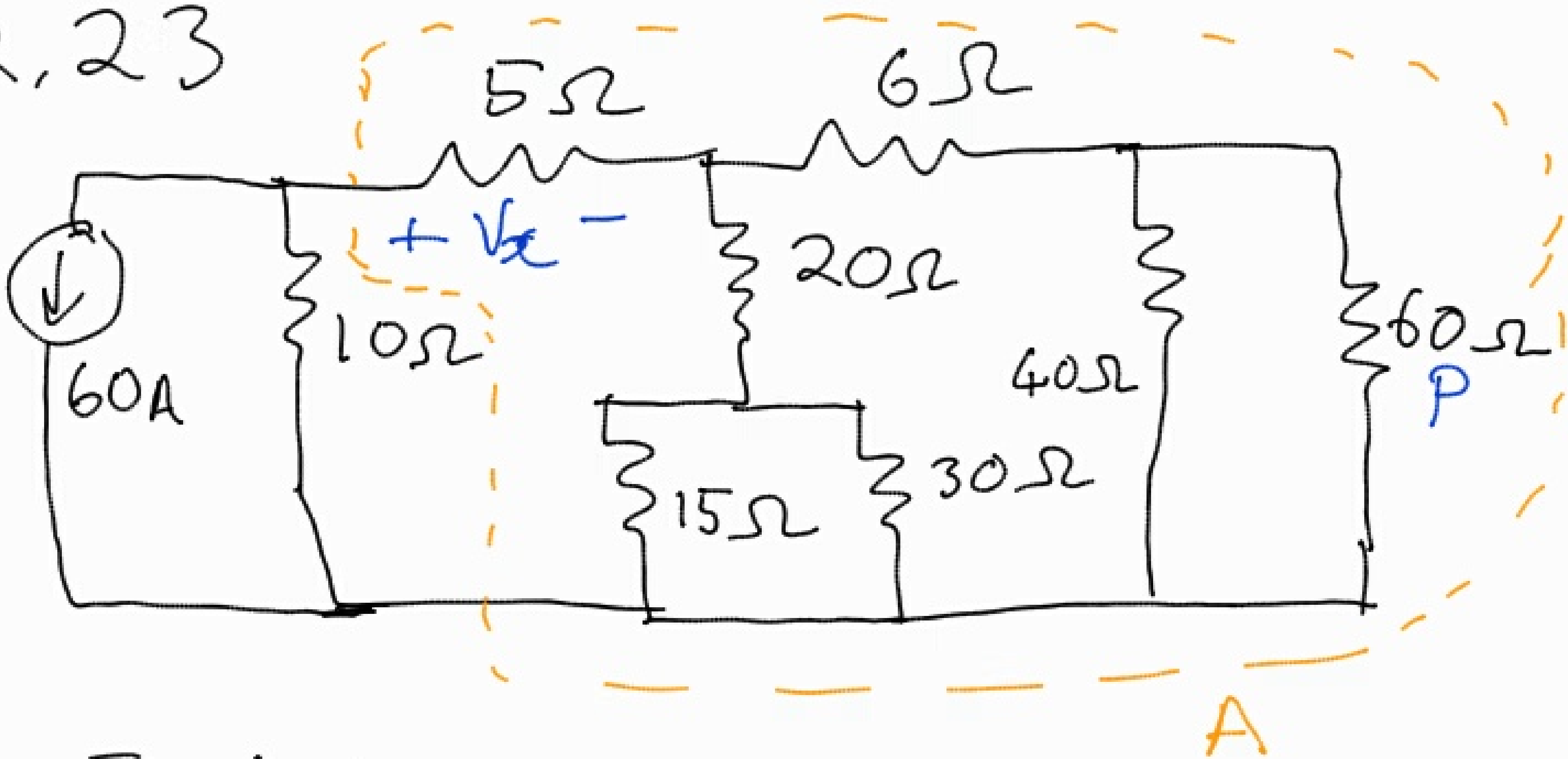
$$V_0 = (-25 - 2V_0) \cdot 10$$

$$V_0 = -250 - 20V_0$$

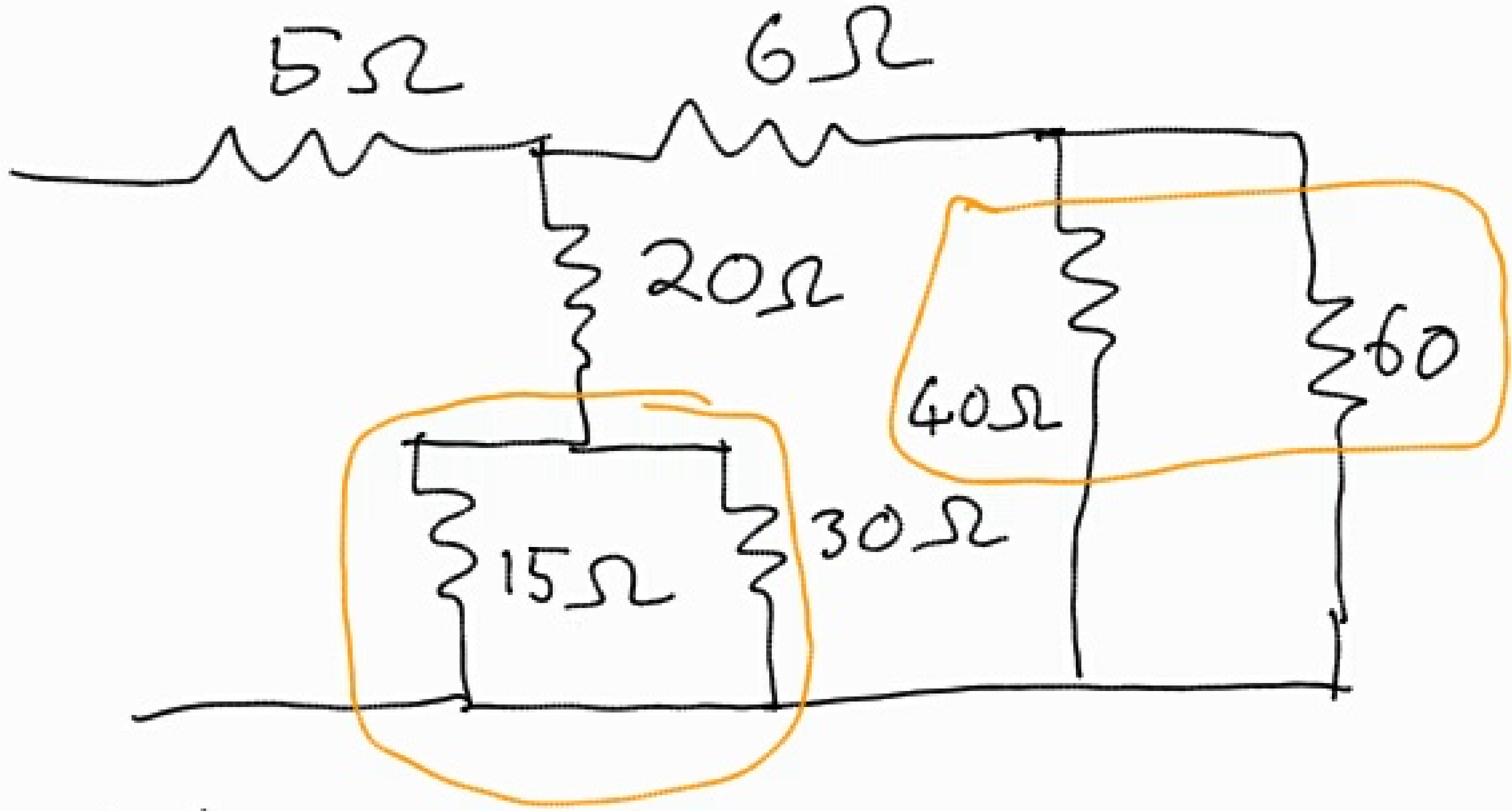
$$21V_0 = -250$$

$$V_0 = 11{,}9\,V$$

2.23



1. Find the current going to part A of the circuit vs the current going to the 10 Ω resistor. By collapssing the resistor network.



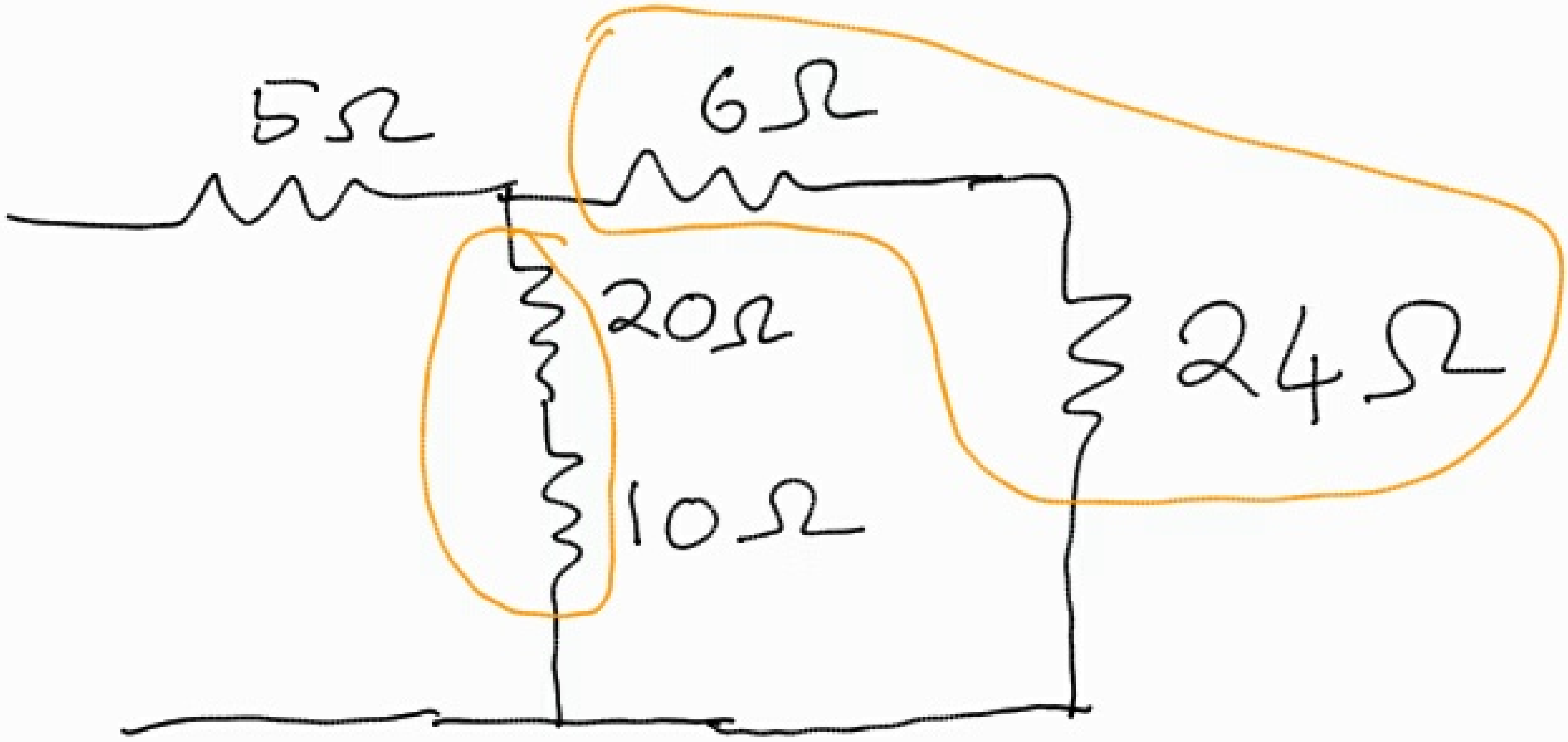
Parallel Resistors

$$15 \| 30 = \frac{15 \times 30}{15 + 30}$$

$$= 10\,\Omega$$

$$40 \| 60 = \frac{40 \times 60}{40 + 60}$$

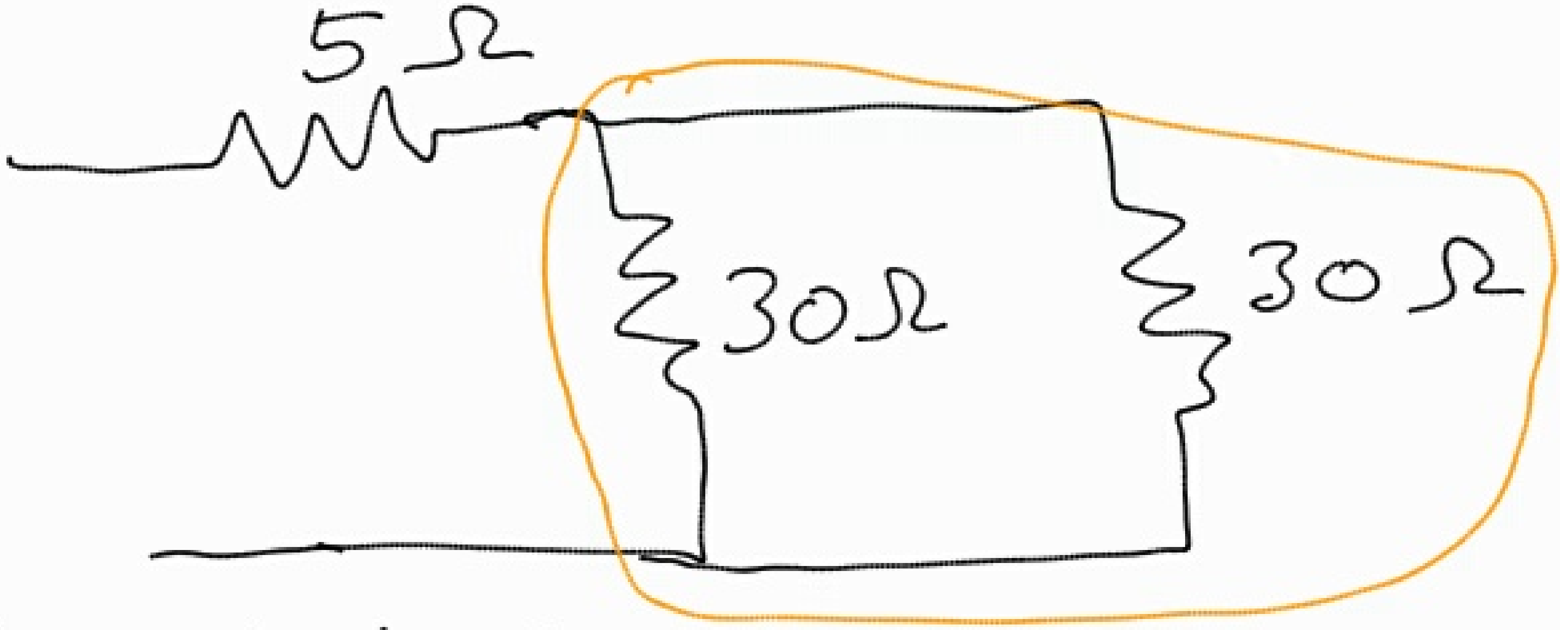
$$= 24\,\Omega$$



Add series resistors.

$20 + 10 = 30\,\Omega$

$6 + 24 = 30\,\Omega$

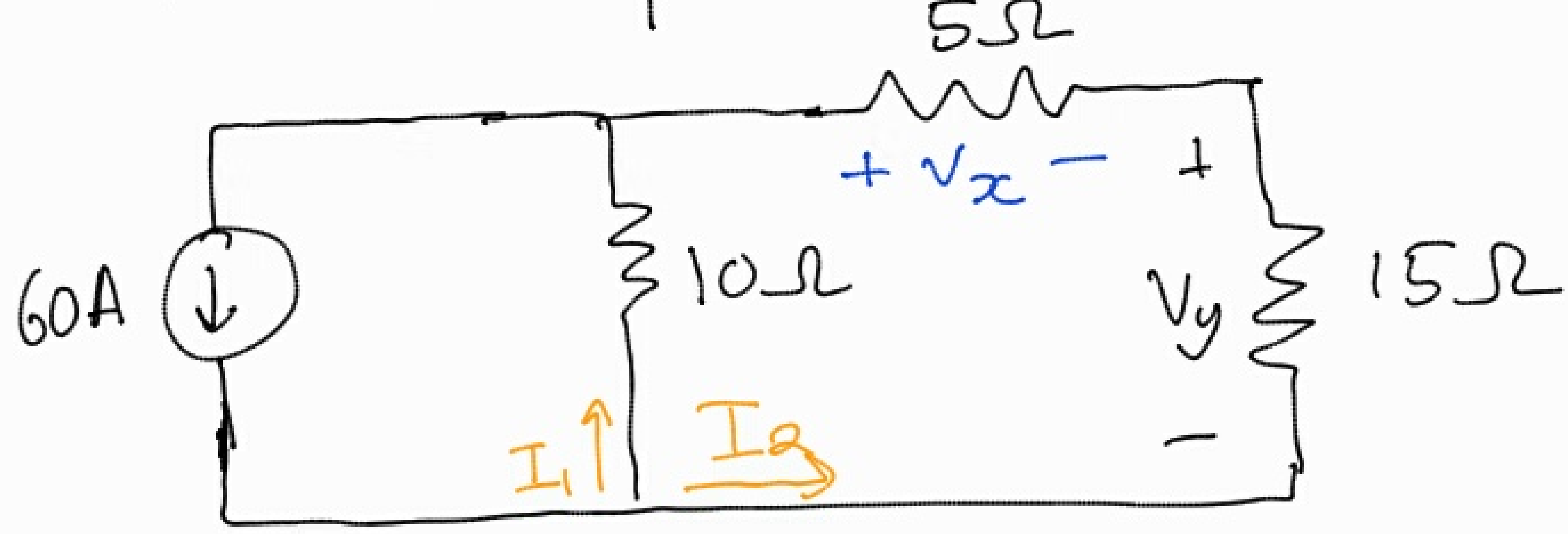


Parallel Resistors

$$30 \| 30 = \frac{30 \times 30}{30 + 30}$$

$$= 15\,\Omega$$

The complete circuit



The [5Ω + 15Ω] pair is in parral with the 10Ω resistor.

Using current division.

$$I_2 = \frac{I \times R_1}{R_1 + R_2}$$

$$I_2 = \frac{60 \times 10}{10 + (5 + 15)}$$

$$= 20 \text{ A}$$

Using Ohm's Law to find V_x

$V = I \cdot R$

$V_x = -20 \times 5$

$V_x = -100\ V$ [Since I_2 enters from negative terminal of V_x]

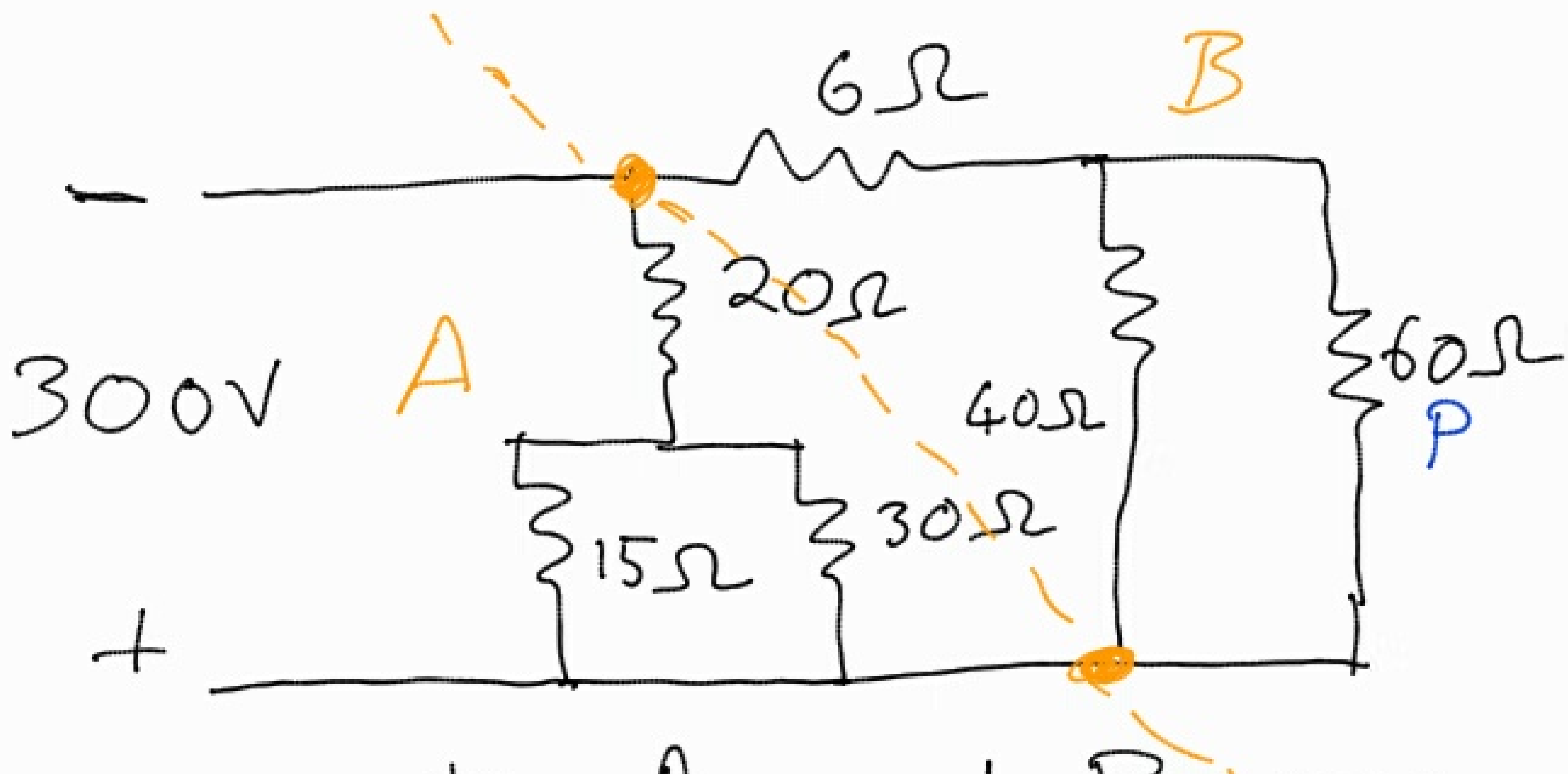
The voltage across the $15\ \Omega$ resistor.

$V = I \cdot R$

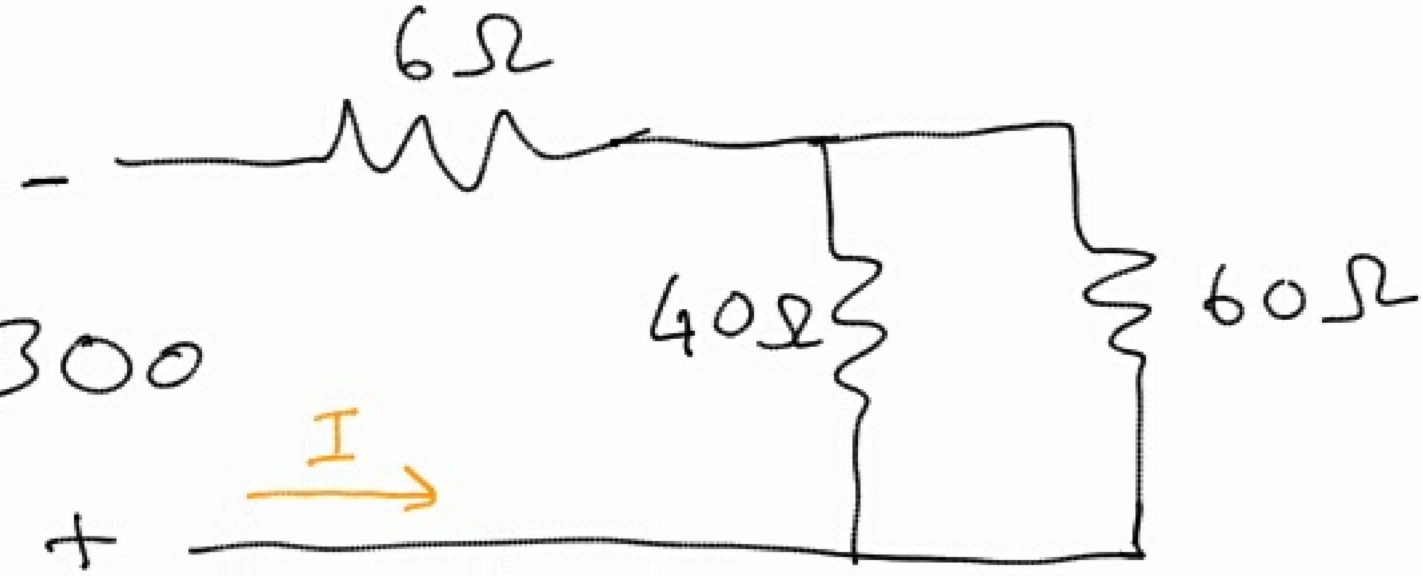
$V_y = -20 \times 15$

$V_y = -300\ V$

To find P expand 15Ω resistor into original network



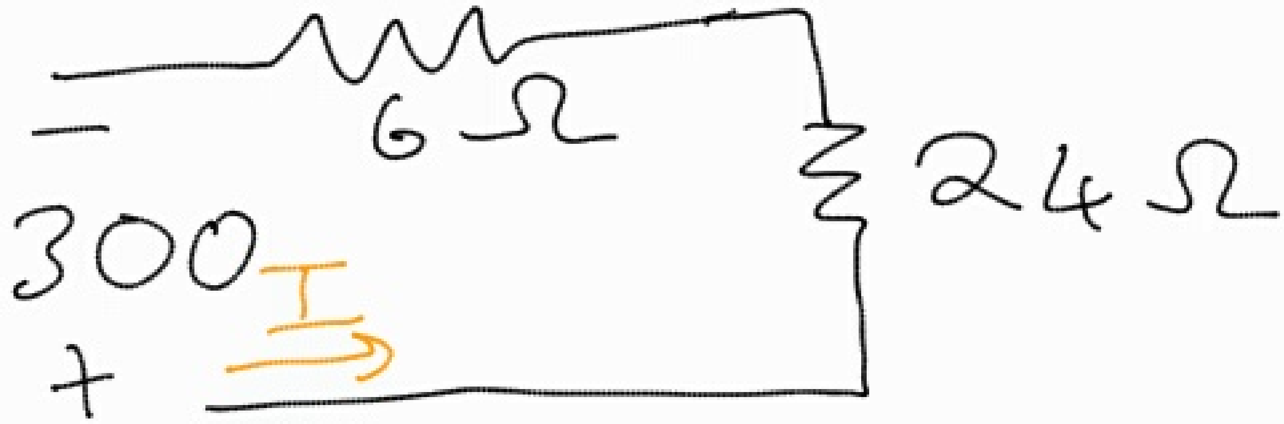
Since parts A and B are parallel, they share the same voltage.



To find I, notice that

$$40 \| 60 = \frac{40 \times 60}{40 + 60}$$

$$= 24$$

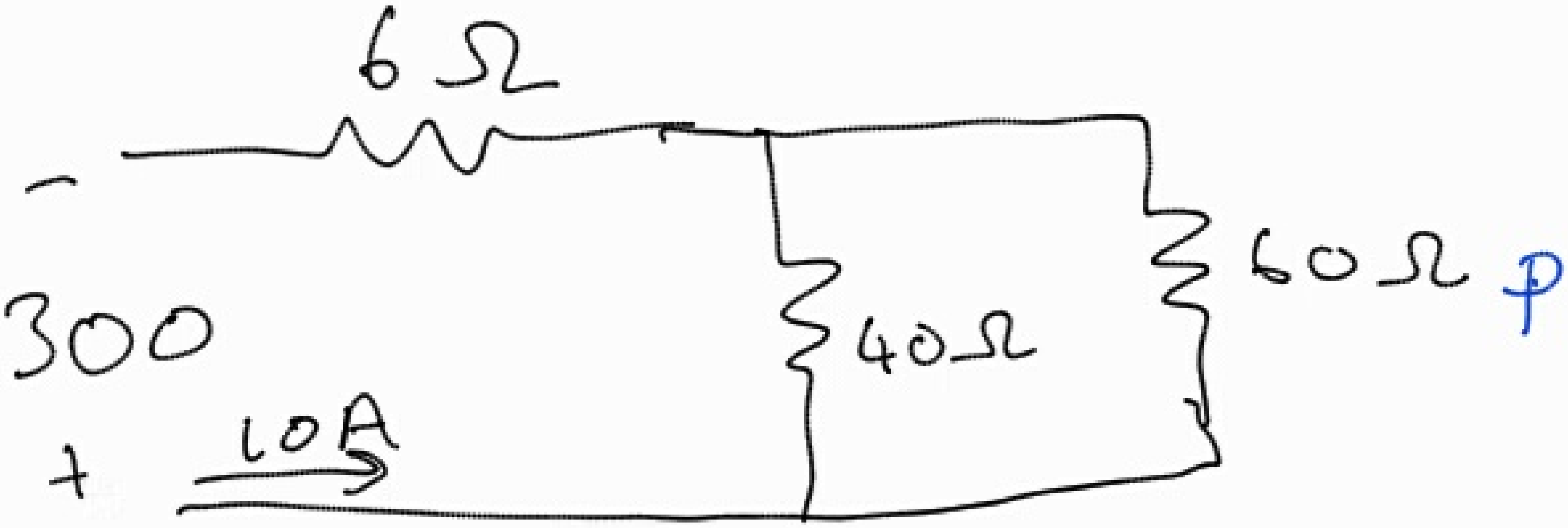


Apply Ohm's Law

$$I = \frac{V}{R}$$

$$I = \frac{300}{30}$$

$$= 10A$$



Finally use current division to find current through 60 Ω

$$I_{60} = \frac{I R_1}{R_1 + R_2}$$

$$I_{60} = \frac{10 \cdot 40}{40 + 60}$$

$$I_{60} = 4 \text{ A}$$

Ohm's law for the voltage

$$V = I . R$$

$$V_{60} = 4 \times 60$$

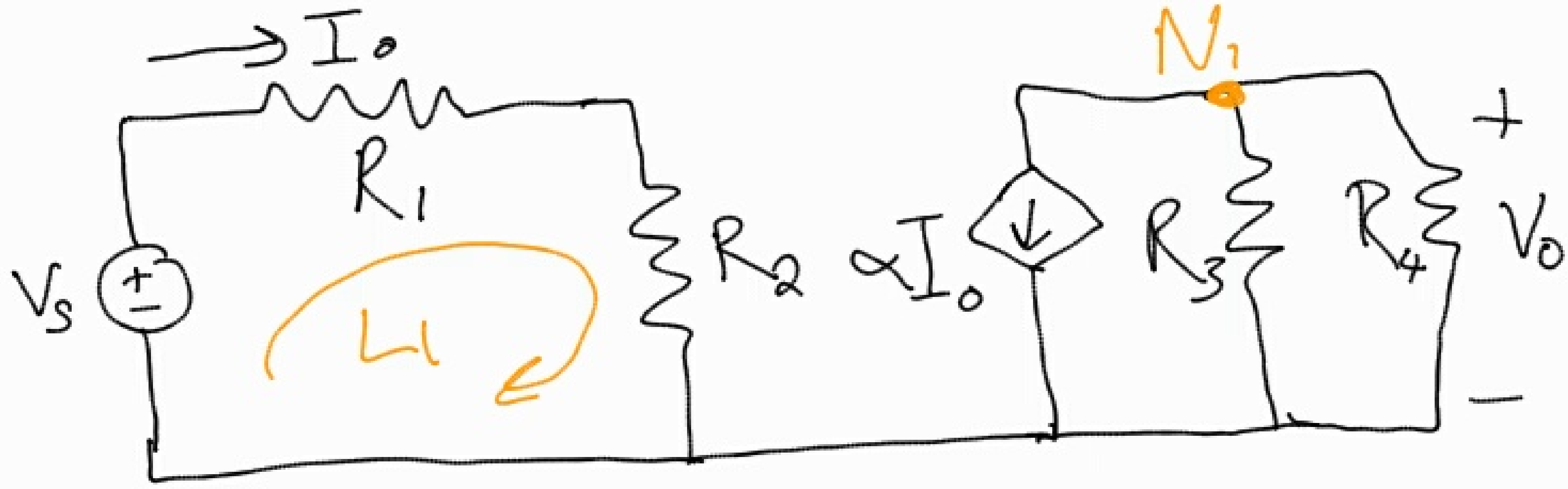
$$V_{60} = 240 \text{ V}$$

$$P = V . I$$

$$= 240 \times 4$$

$$= 960 \text{ W}$$

2,24



Apply KVL at loop 1. $(\sum v = 0)$

$$-V_s + I_o R_1 + I_o R_1 = 0 \quad (V = IR)$$

$$I_o(R_1 + R_2) = V_s$$

$$I_o = \frac{V_s}{R_1 + R_2} \quad (1)$$

Apply KCL at node 1 $(\sum I = 0)$

$$\alpha I_o + \frac{V_o}{R_3} + \frac{V_o}{R_4} = 0 \quad \left(I = \frac{V}{R}\right)$$

$$\alpha I_o = -V_o\left(\frac{1}{R_3} + \frac{1}{R_4}\right)$$

$$I_o = \frac{-V_o}{\alpha}\left(\frac{1}{R_3} + \frac{1}{R_4}\right) \quad (2)$$

setting ① = ②

$$\frac{V_s}{R_1+R_2} = \frac{-V_o}{\alpha}\left(\frac{1}{R_3}+\frac{1}{R_4}\right)$$

$$\frac{V_s}{R_1+R_2} = \frac{-V_o}{\alpha}\left(\frac{R_4+R_3}{R_3R_4}\right)$$

$$\frac{-\alpha R_3 R_4}{R_4+R_3}\cdot\left(\frac{1}{R_1+R_2}\right) = \frac{V_o}{V_s}$$

---

let $R_1 = R_2 = R_3 = R_4 = R$

$$\frac{-\alpha R.R}{(R+R)(R+R)} = \frac{V_o}{V_s}$$

$$\frac{-\alpha R^2}{2R.2R} = \frac{V_o}{V_s}$$

$$\frac{-\alpha R^2}{4R^2} = \frac{V_o}{V_s}$$

$$\frac{-\alpha}{4} = \frac{V_o}{V_s}$$

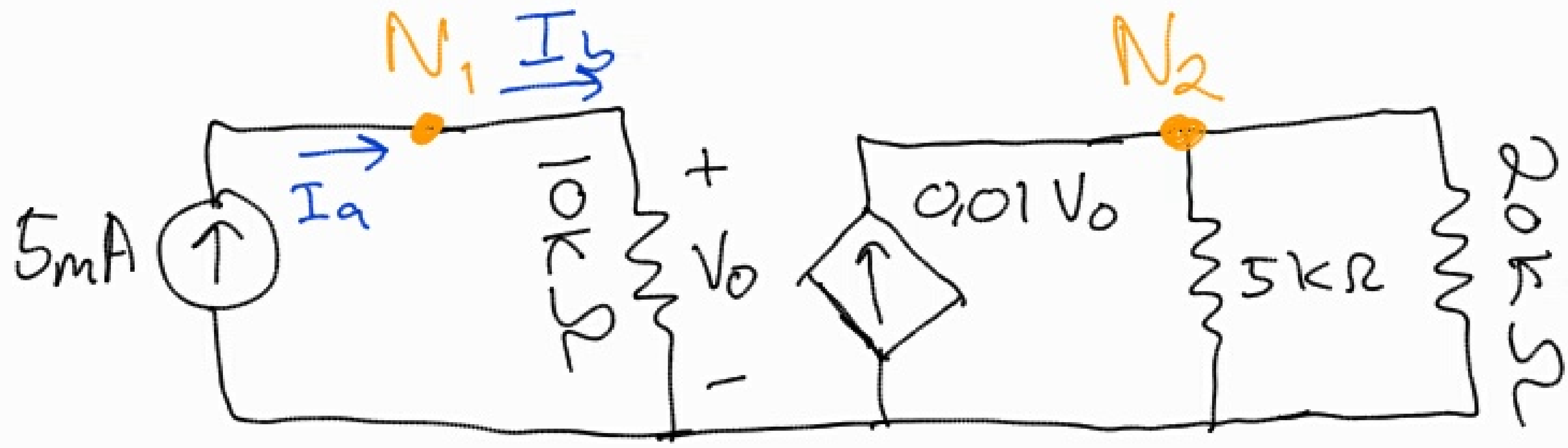
let $\left|\frac{V_o}{V_s}\right| = 10$

$$\left|\frac{\alpha}{-4}\right| = \left|\frac{V_o}{V_s}\right|$$

$$\frac{\alpha}{4} = 10$$

$$\alpha = 40$$

2.25



Apply KCL and Ohm's Law at node 1, $\sum I = 0$, $I = \frac{V}{R}$

$$I_a - I_b = 0$$

$$5\times10^{-3} - \frac{V_o}{10\times10^{3}} = 0$$

$$\frac{-V_o}{10\times10^{3}} = -5\times10^{-3}$$

$$V_o = (-5\times10^{-3})(-10\times10^{3})$$

$$V_o = 50\ V$$

Apply current division at node 2.

$$I_2 = \frac{I \cdot R_1}{R_1 + R_2}$$

$$I = \frac{0,01 V_0 \times 5\times10^3}{5\times10^3 + 20\times10^3}$$

$$I = \frac{0,01\times50\times5\times10^3}{5\times10^3 + 20\times10^3}$$

$$I = 0,1 \text{ A}$$

$$I = 100 \text{ mA}$$

Ohm's Law

$$V = IR$$

$$V = 0,1 \times 20\times10^3$$

$$V = 2000 \text{ V}$$

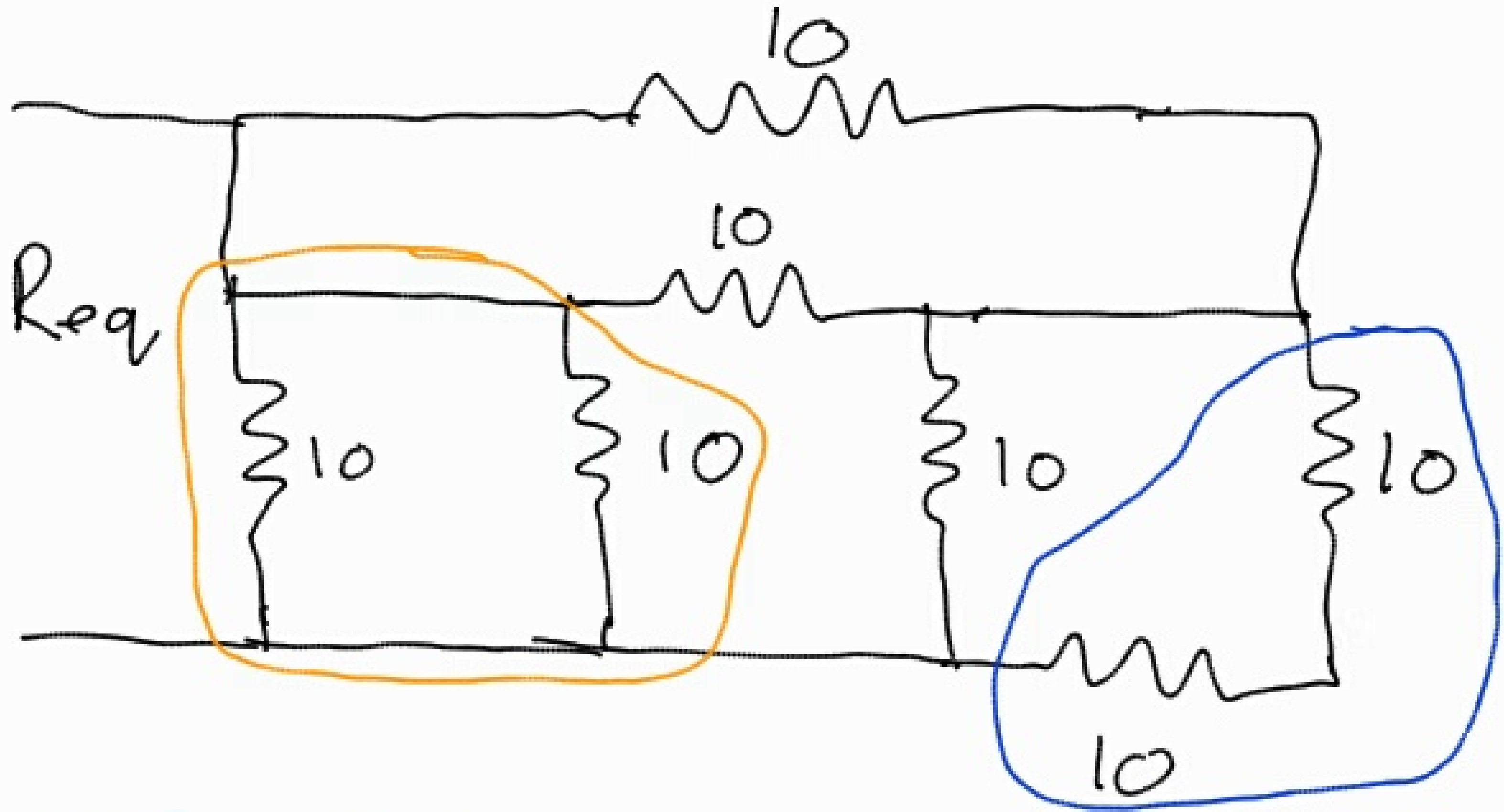
$$V = 2 \text{ kV}$$

$$P = V \cdot I$$

$$P = 2000 \times 0{,}1$$

$$P = 200 \text{ W}$$

2,29



- Series

$$R_{eq} = R_1 + R_2$$

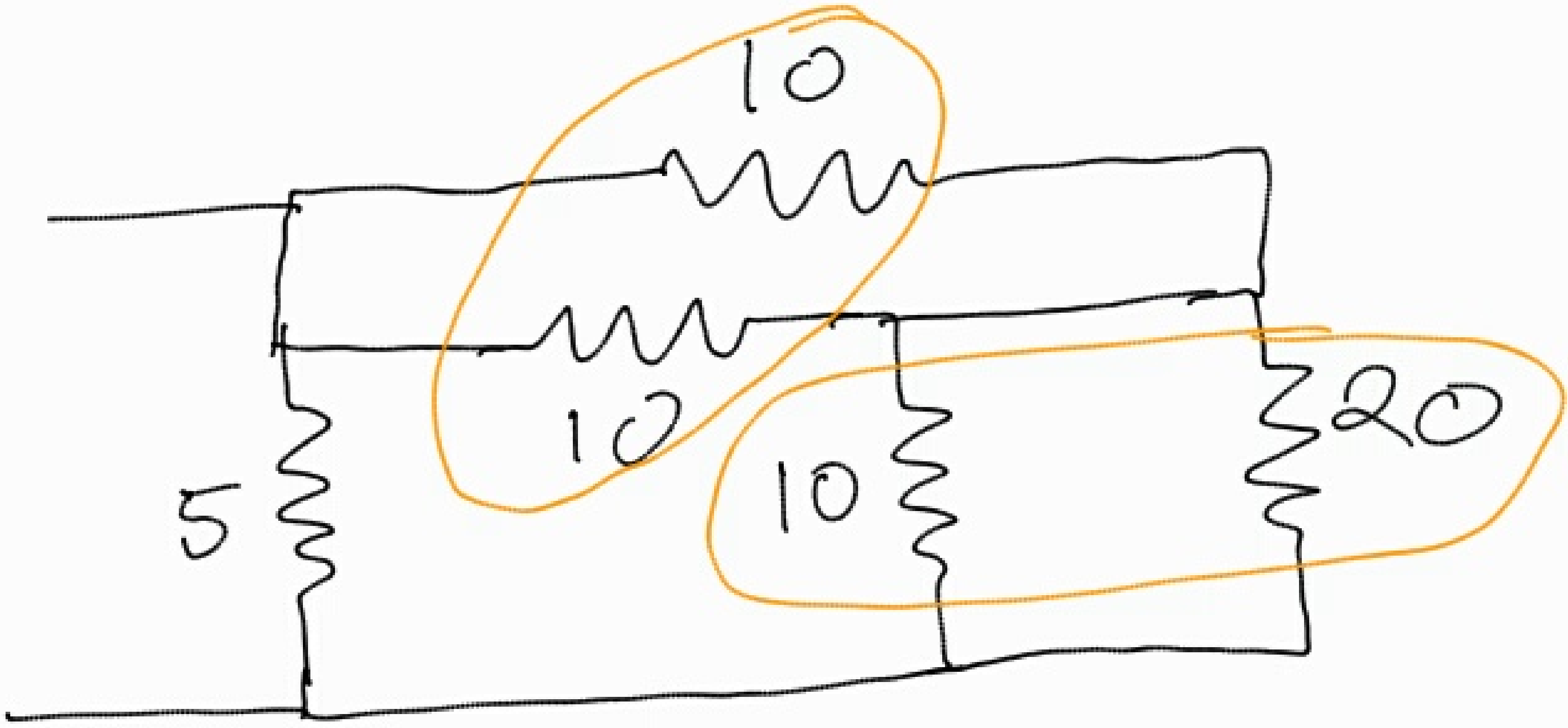
- Parallel

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$10 + 10 = 20\,\Omega$$

$$10 \parallel 10 = \frac{10 \times 10}{10 + 10}$$

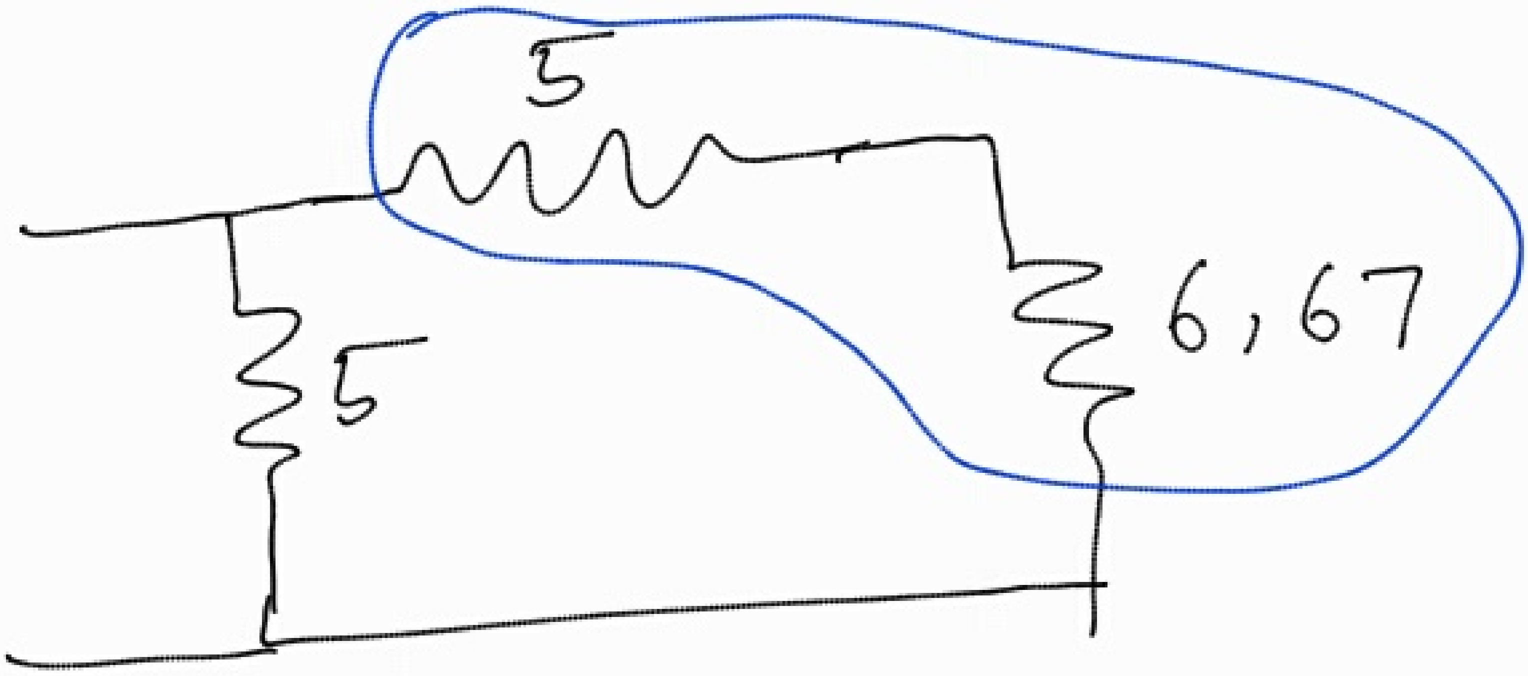
$$= 5\,\Omega$$



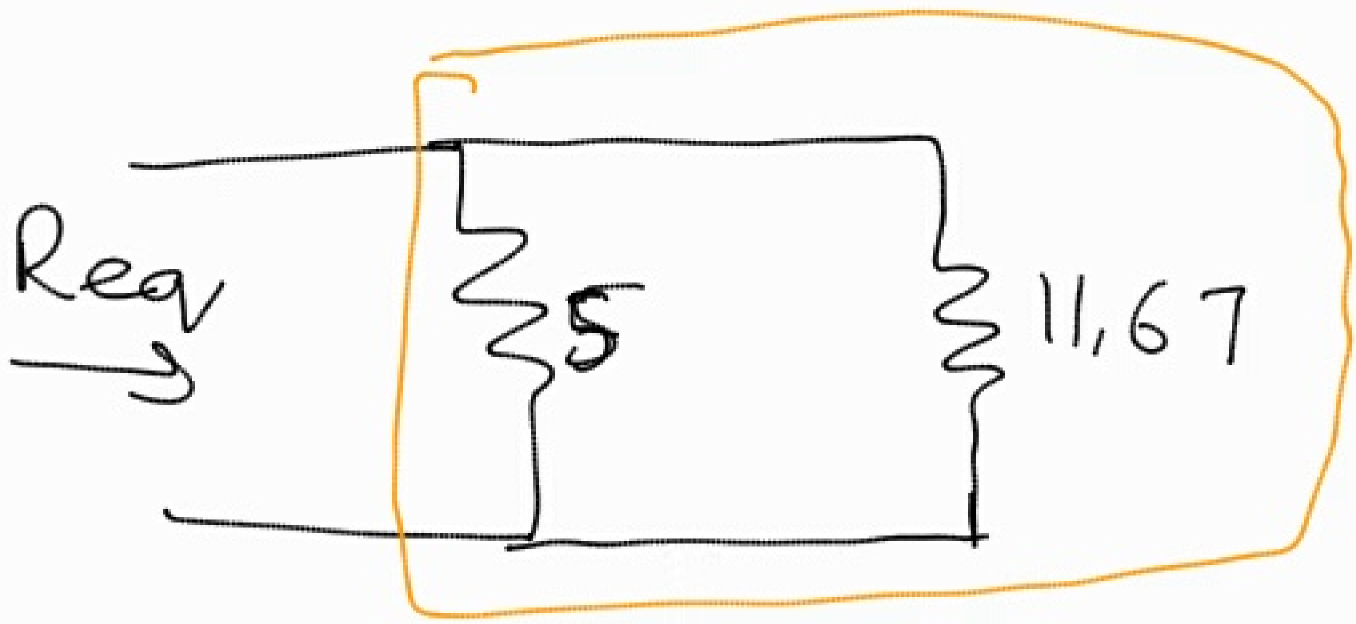
$$10 \parallel 20 = \frac{10 \times 20}{10 + 20}$$

$$= \frac{20}{3} = 6{,}67\,\Omega$$

$$10 \parallel 10 = 5\,\Omega$$



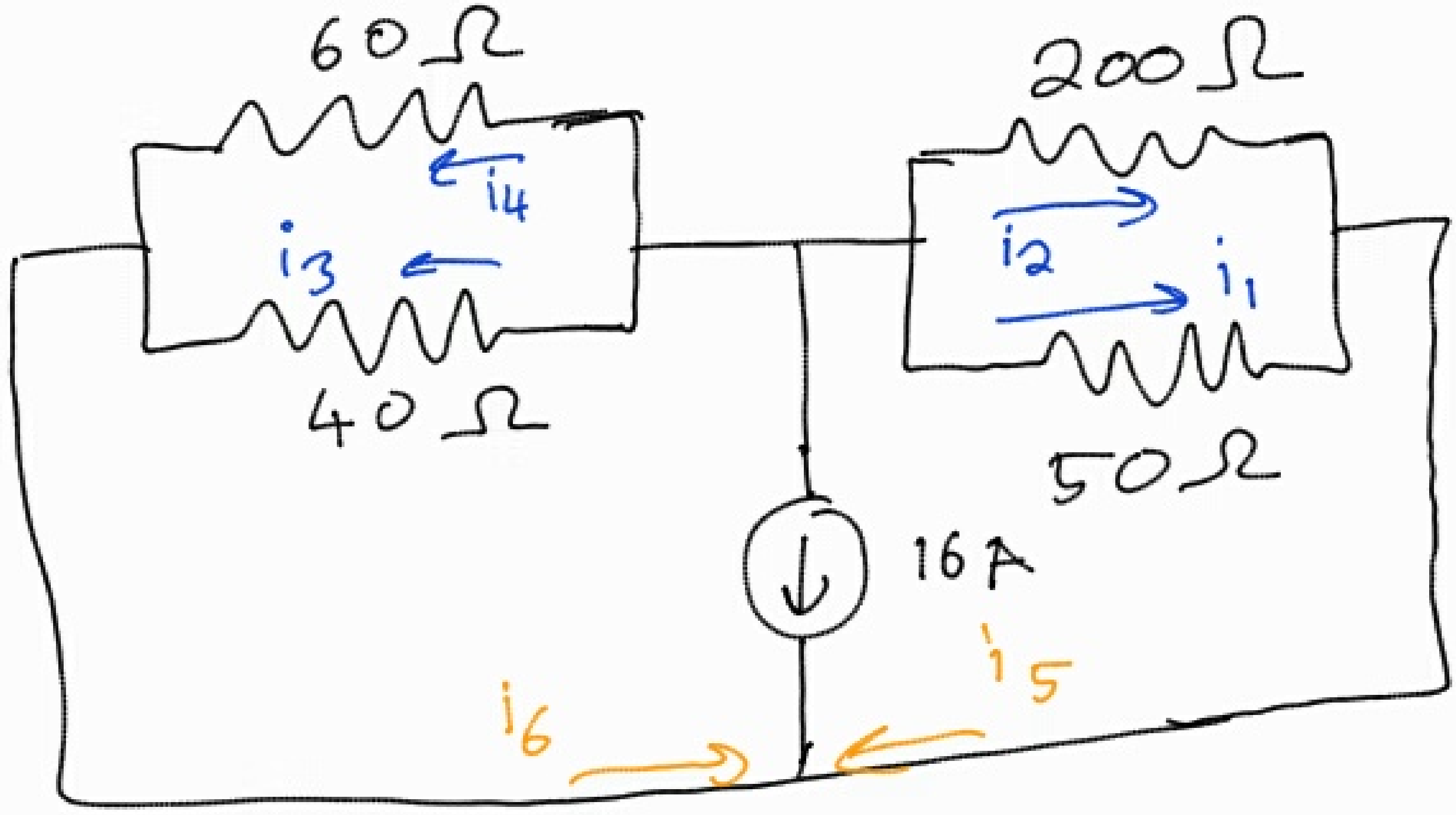
$$5 + 6{,}67 = 11{,}67\ \Omega$$



$$5 \parallel 11{,}67 = \frac{5 \times 11{,}67}{5 + 11{,}67}$$

$$= 3{,}50\ \Omega$$

2,32



let $R_a = 60 \| 40$

$$= \frac{40 \times 60}{60 + 40}$$

$$= 24\ \Omega$$

$R_b = 200 \| 50$

$$= \frac{200 \times 50}{200 + 50}$$

$$= 40\ \Omega$$

Apply current division to find i_5 & i_6.

- Note $16 + i_5 + i_6 = 0$

$$16 = -i_5 - i_6$$

i_5 & i_6 are negative

$$I_2 = I \cdot \frac{R_1}{R_1 + R_2}$$

$$i_5 = -16 \times \frac{24}{24 + 40}$$

$$= -6 \text{ A}$$

$$i_6 = -16 \times \frac{40}{24 + 40}$$

$$= -10 \text{ A}$$

$$i_1 = i_5 \times \frac{200}{200+50}$$

$$= -6 \times \frac{200}{200+50}$$

$$= -4.8 \text{ A}$$

$$i_2 = i_5 \times \frac{50}{200+50}$$

$$= -6 \times \frac{50}{200+50}$$

$$= -1.2 \text{ A}$$

$$i_3 = i_6 \times \frac{60}{40+60}$$

$$= -10 \times \frac{60}{40+60}$$

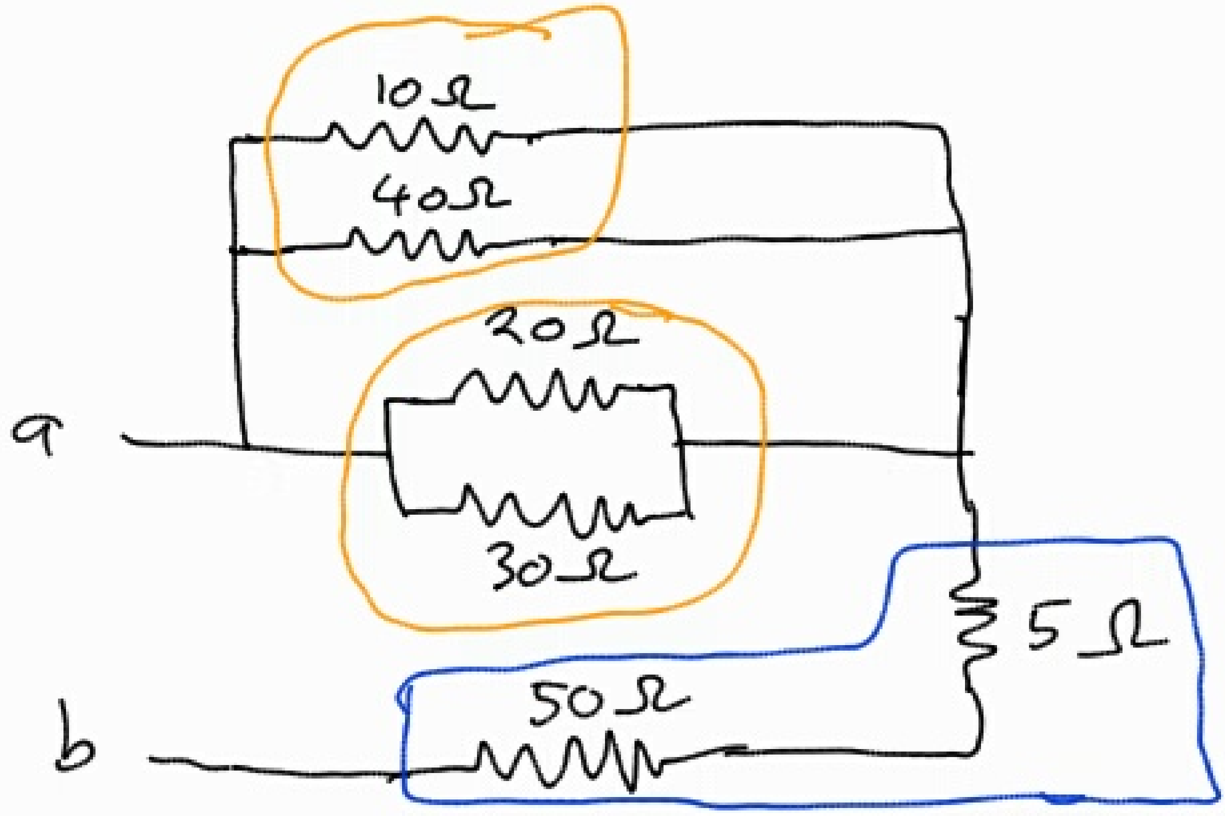
$$= -6 \text{ A}$$

$$i_4 = i_6 \times \frac{40}{40+60}$$

$$= -10 \times \frac{40}{40+60}$$

$$= -4 \text{ A}$$

2.45



- Series

$$R_{eq} = R_1 + R_2$$

- Parallel

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

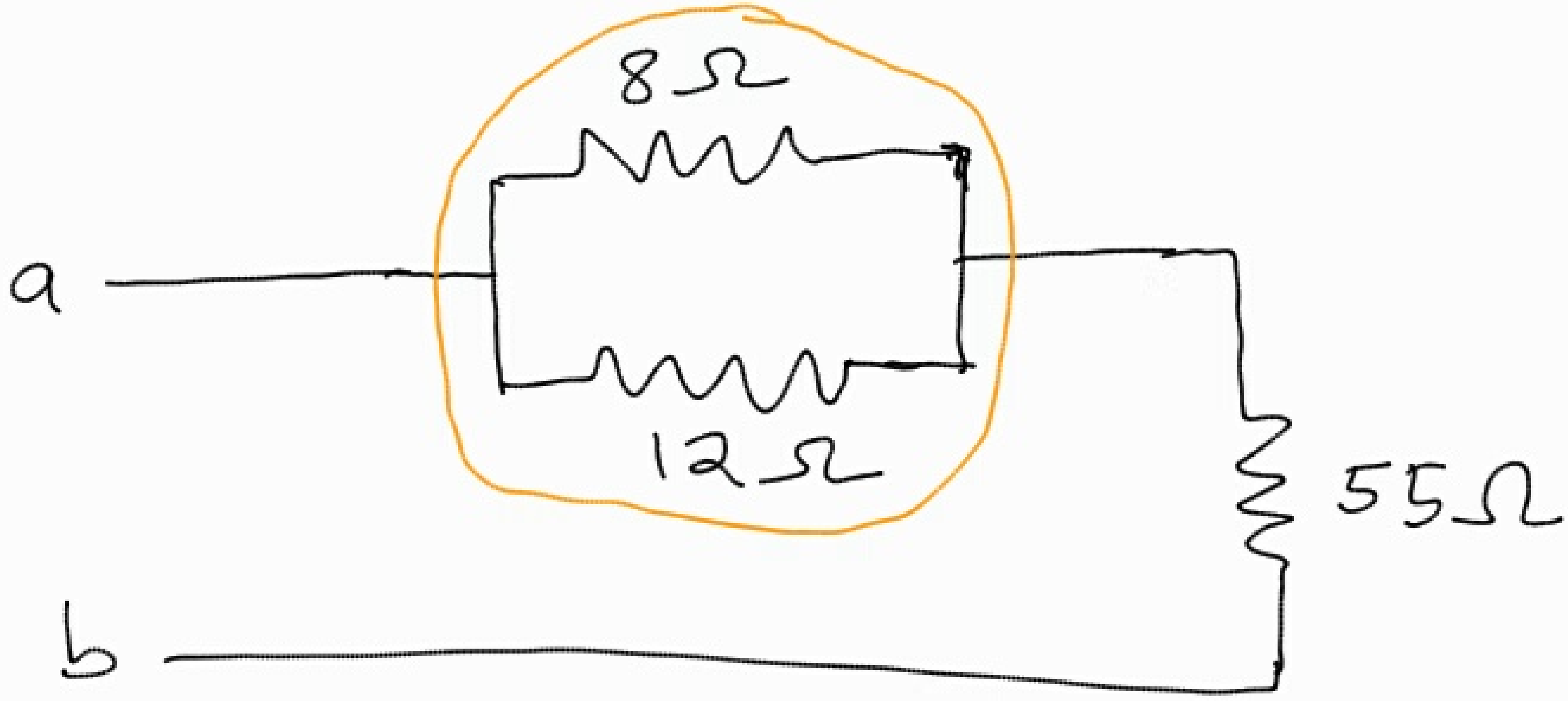
$$10 \| 40 = \frac{10 \times 40}{10 + 40}$$

$$= 8\,\Omega$$

$$20 \| 30 = \frac{20 \times 30}{20 + 30}$$

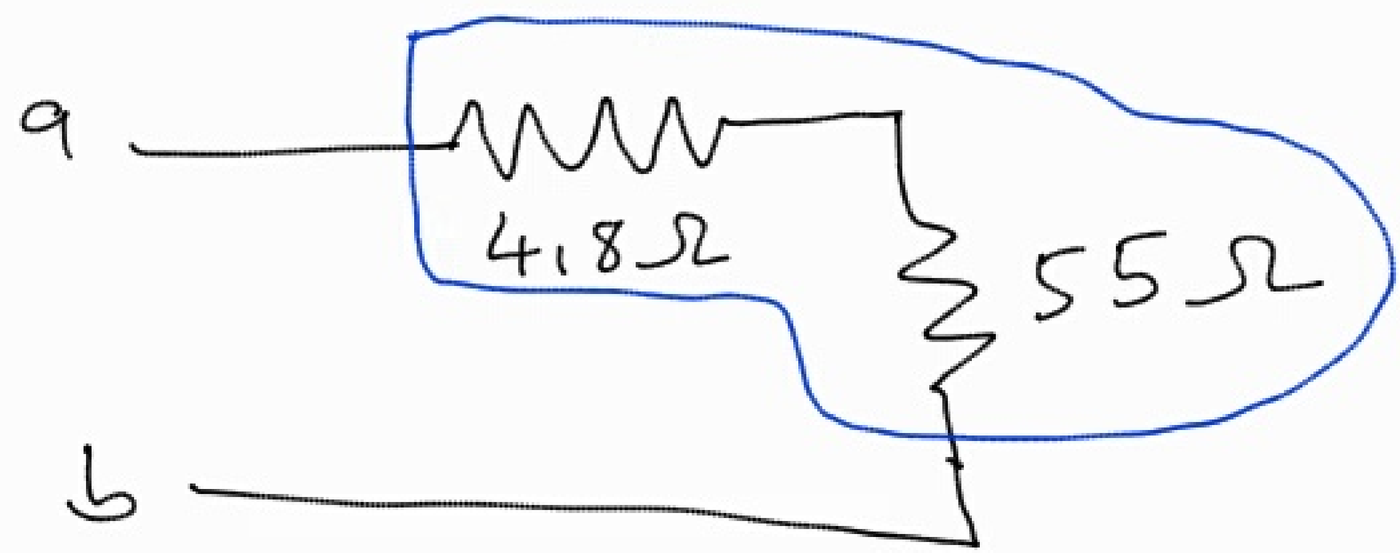
$$= 12$$

$$5 + 50 = 55\,\Omega$$

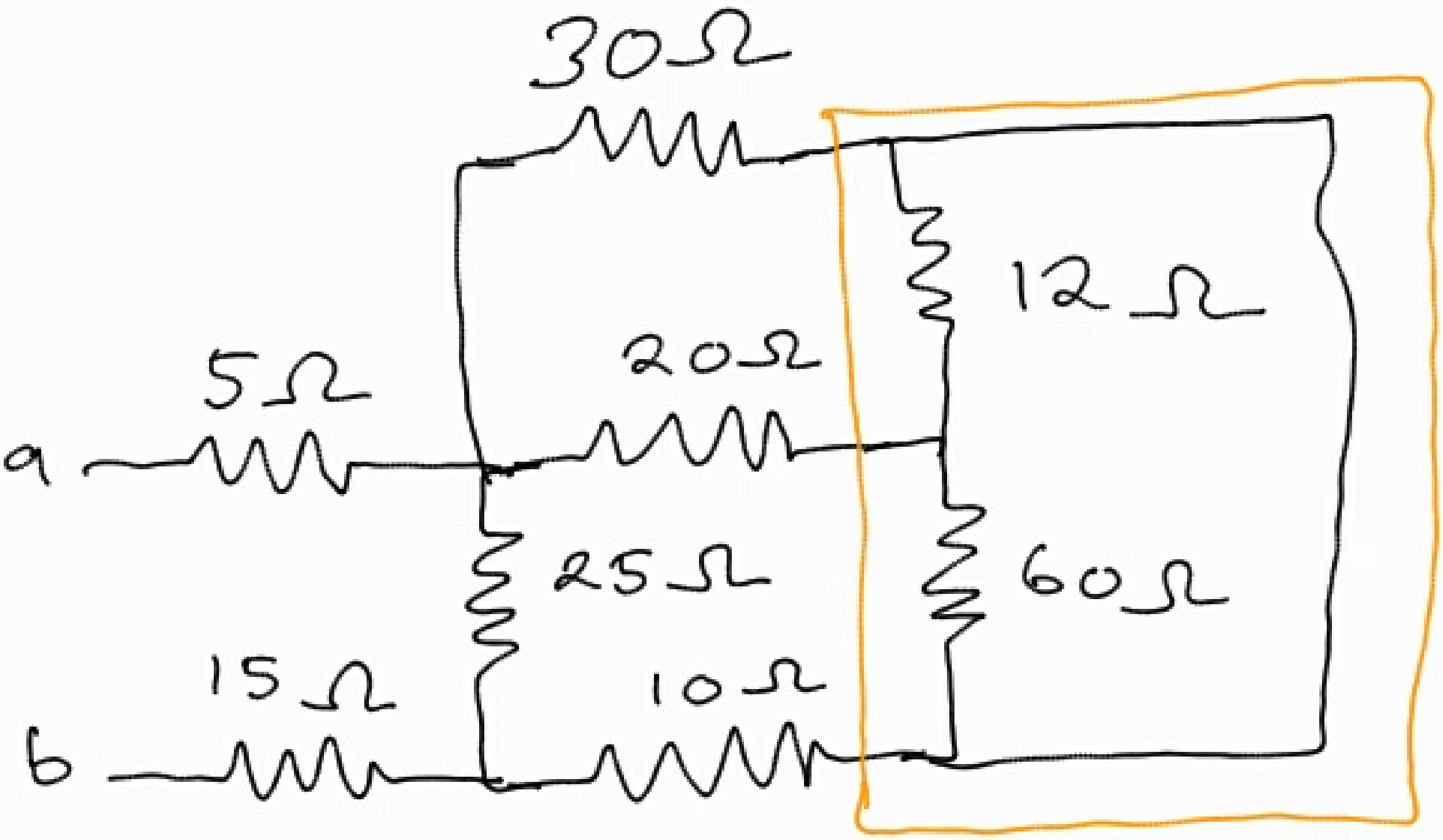


$$8 \| 12 = \frac{8 \times 12}{8+12}$$

$$= 4{,}8\ \Omega$$

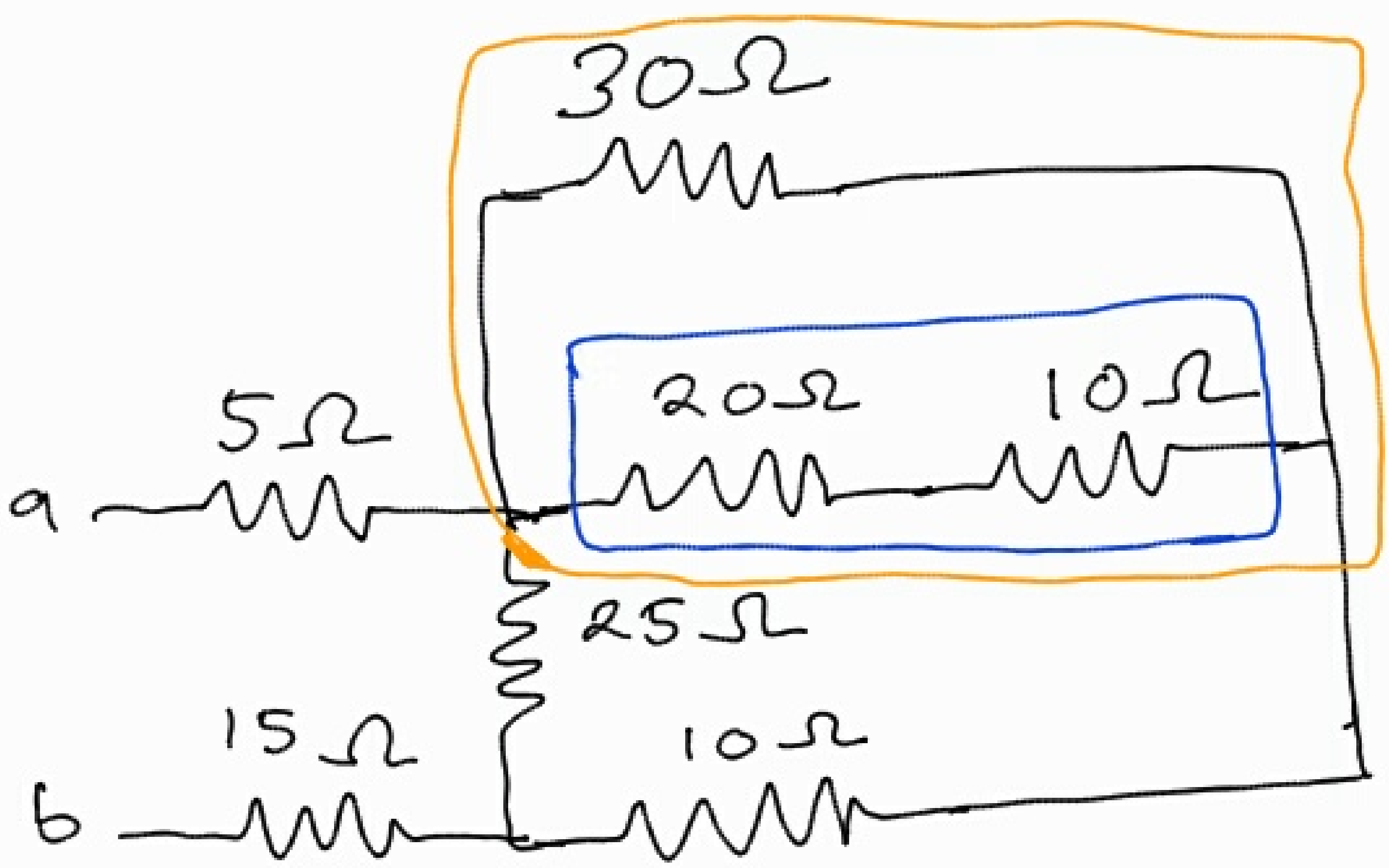


$$4{,}8 + 55 = 59{,}8\ \Omega$$



$$12 \| 60 = \frac{12 \times 60}{12 + 60}$$

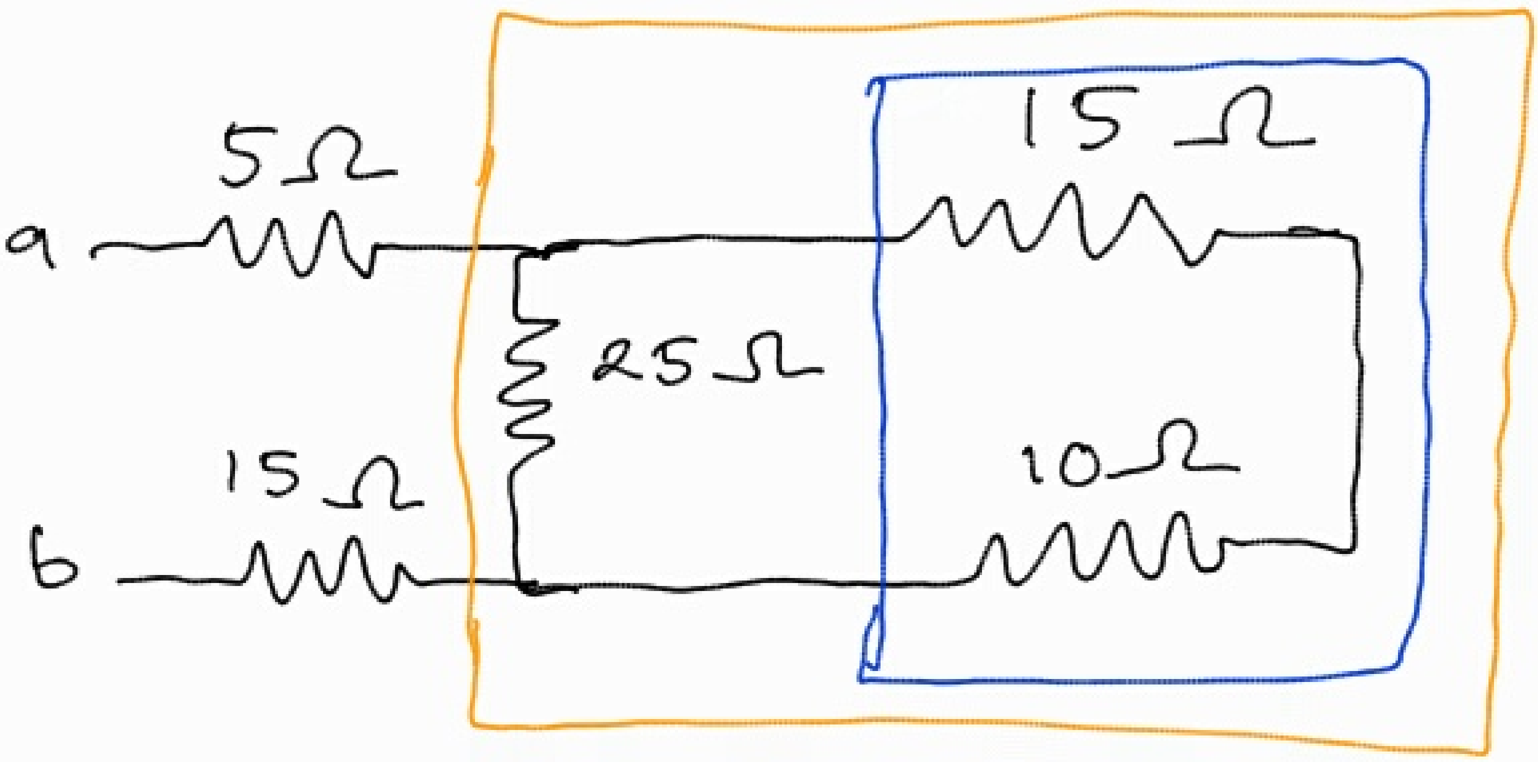
$$= 10\,\Omega$$



$$10 + 20 = 30\,\Omega$$

$$30 \parallel 30 = \frac{30 \times 30}{30 + 30}$$

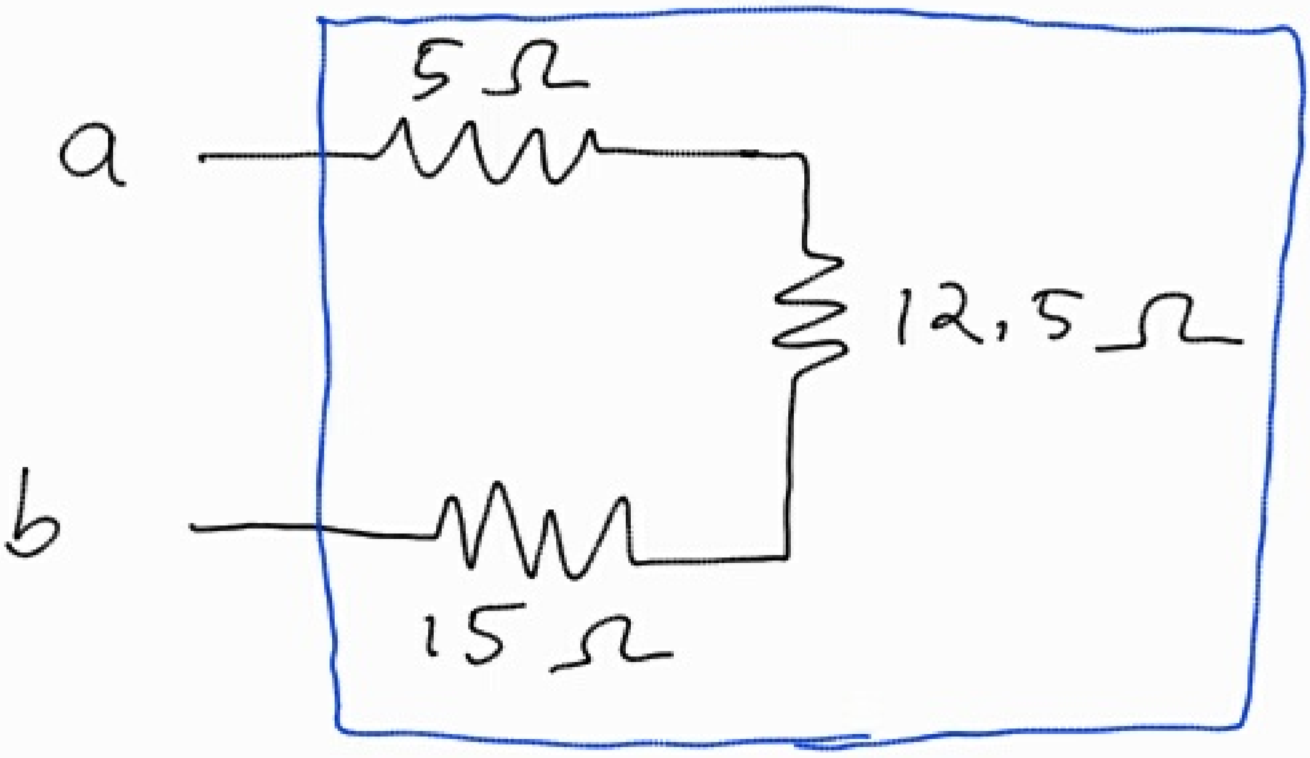
$$= 15\,\Omega$$



$$10 + 15 = 25\,\Omega$$

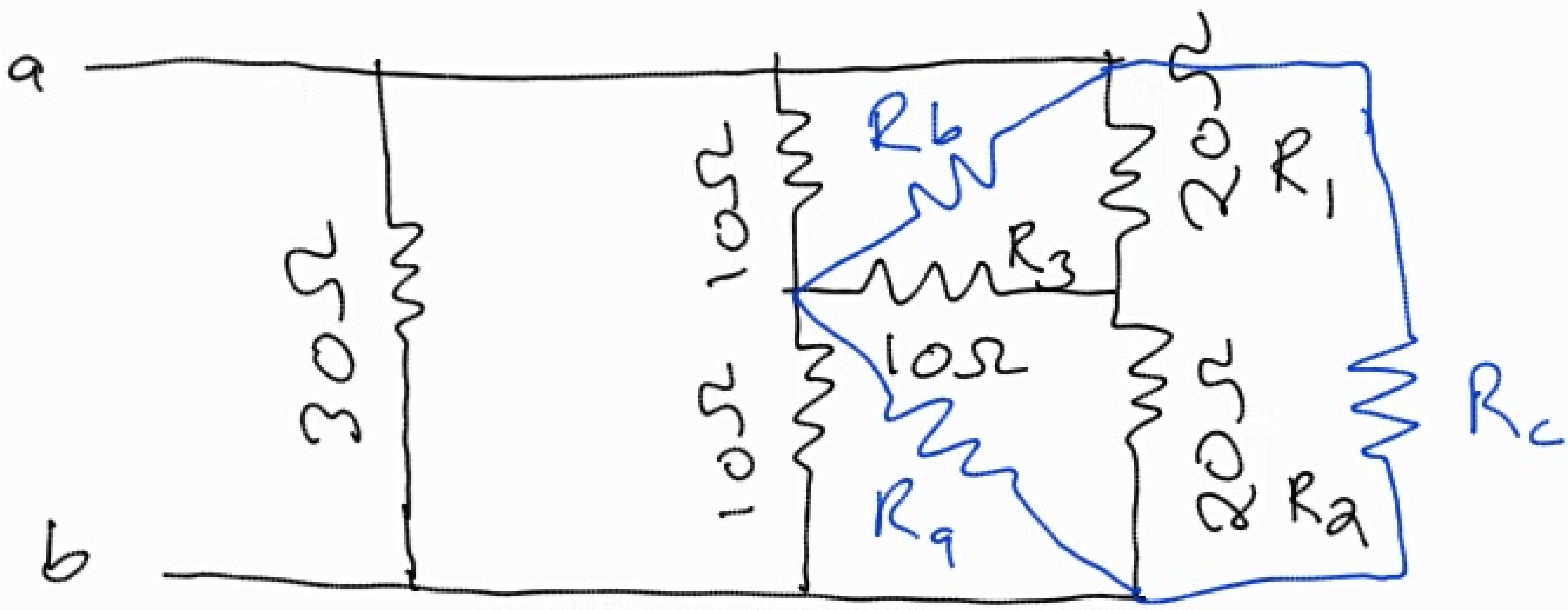
$$25 \parallel 25 = \frac{25 \times 25}{25 + 25}$$

$$= 12{,}5\,\Omega$$



$$R_{eq} = 5 + 12{,}5 + 15$$
$$= 32{,}5\ \Omega$$

2.51



$$S = R_1R_2 + R_2R_3 + R_3R_1$$

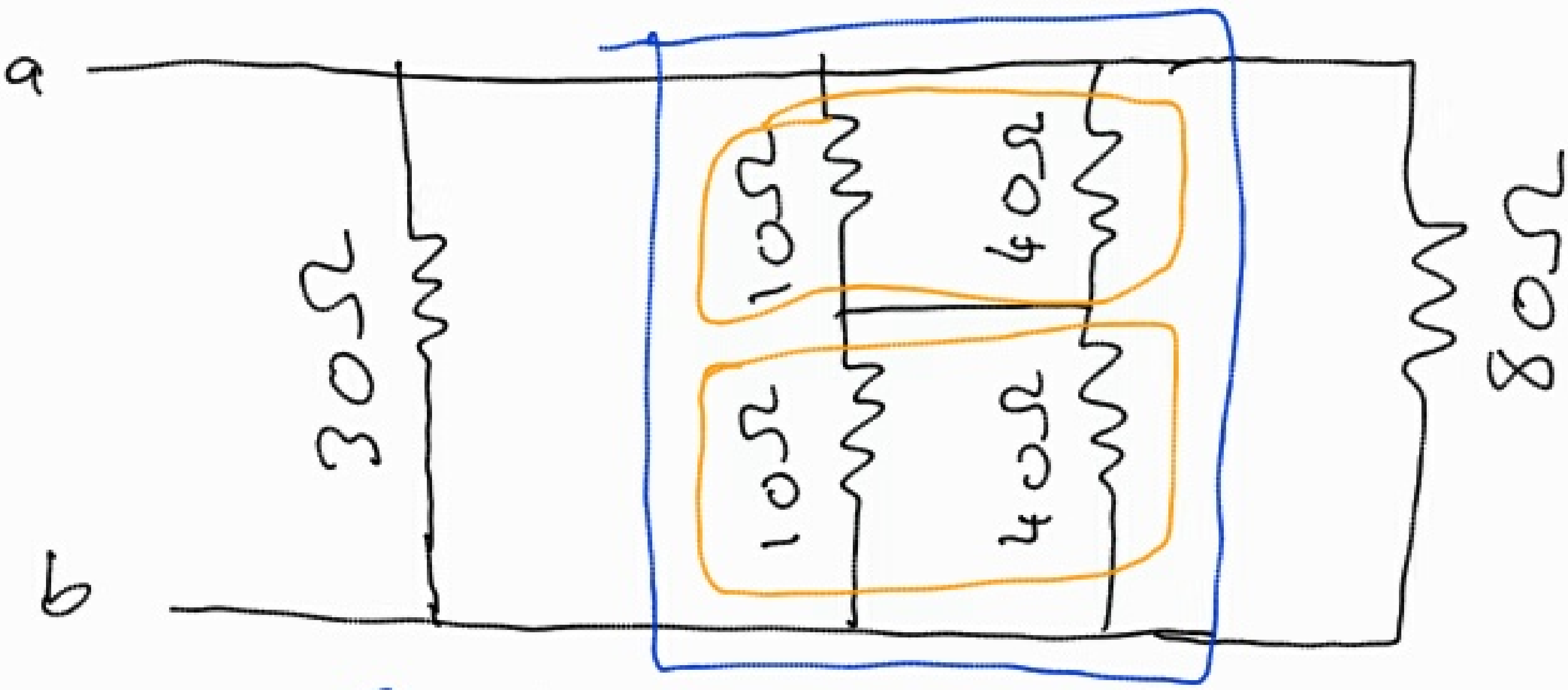
$$= 20 \times 20 + 20 \times 10 + 10 \times 20$$

$$= 800\ \Omega$$

$$R_a = \frac{S}{R_1} = \frac{800}{20} = 40\Omega$$

$$R_b = \frac{S}{R_2} = \frac{800}{20} = 40\Omega$$

$$R_c = \frac{S}{R_3} = \frac{800}{10} = 80\Omega$$



- Series

$$R_{eq} = R_1 + R_2$$

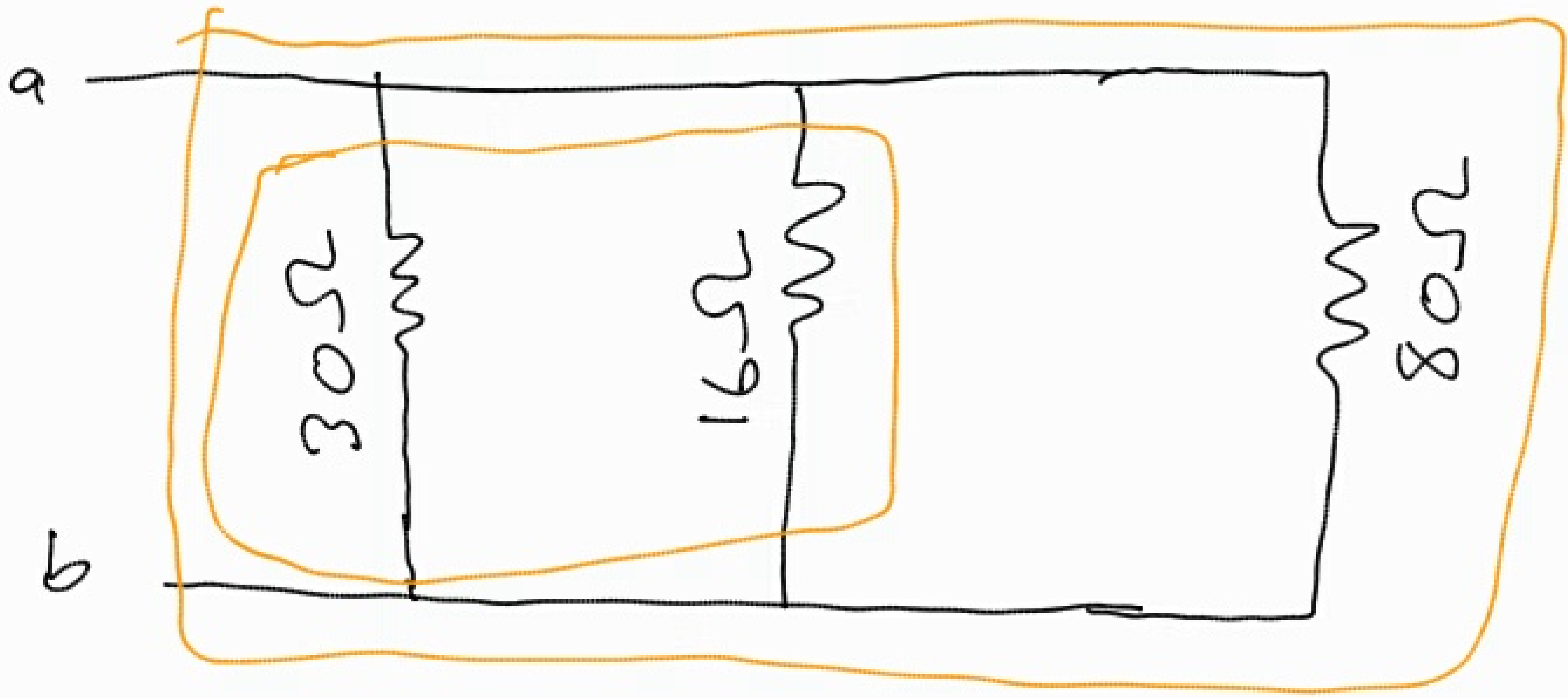
- Parallel

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$10 \| 40 = \frac{10 \times 40}{10 + 40}$$

$$= 8\,\Omega$$

$$8 + 8 = 16\,\Omega$$

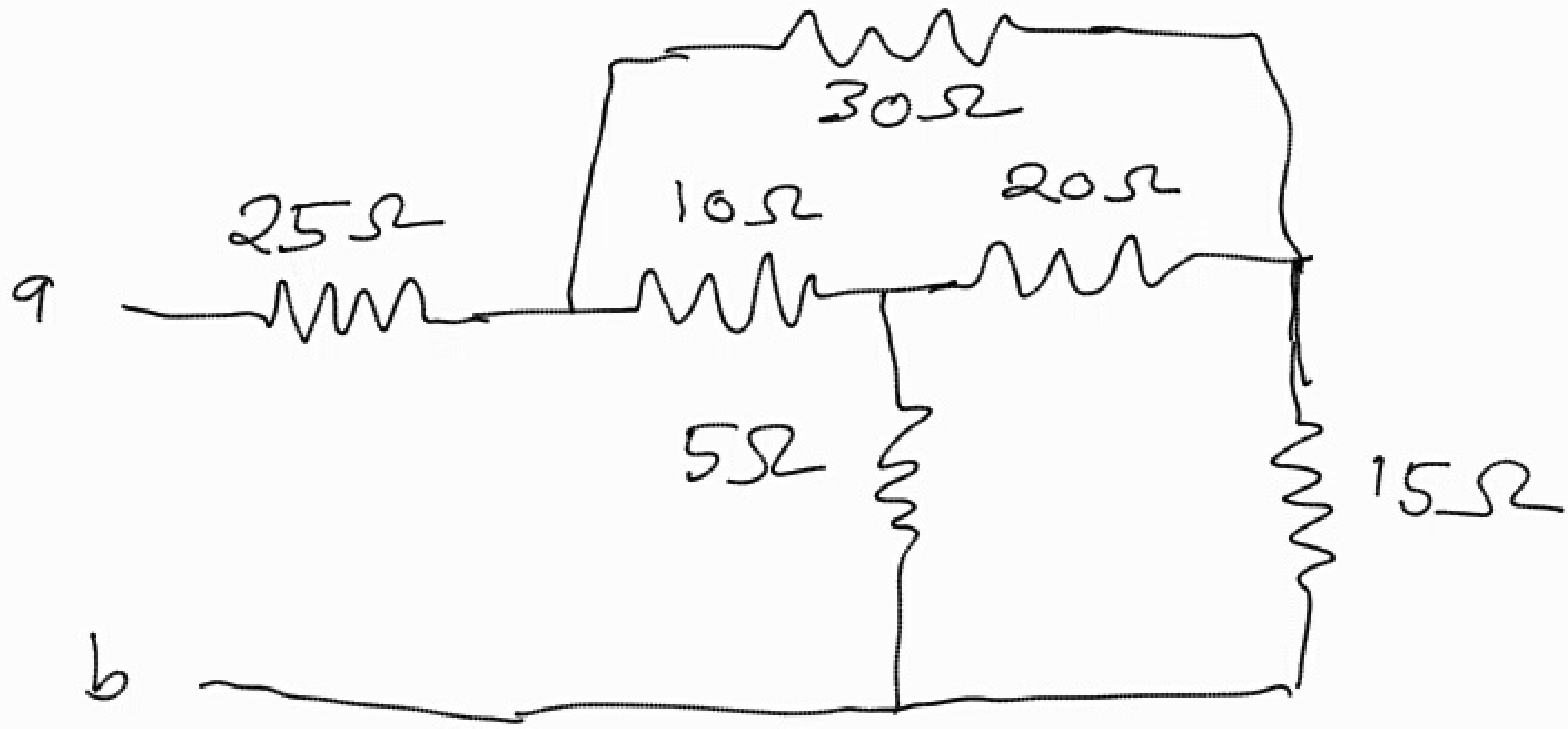


$$16 \parallel 30 = \frac{16 \times 30}{16 + 30}$$

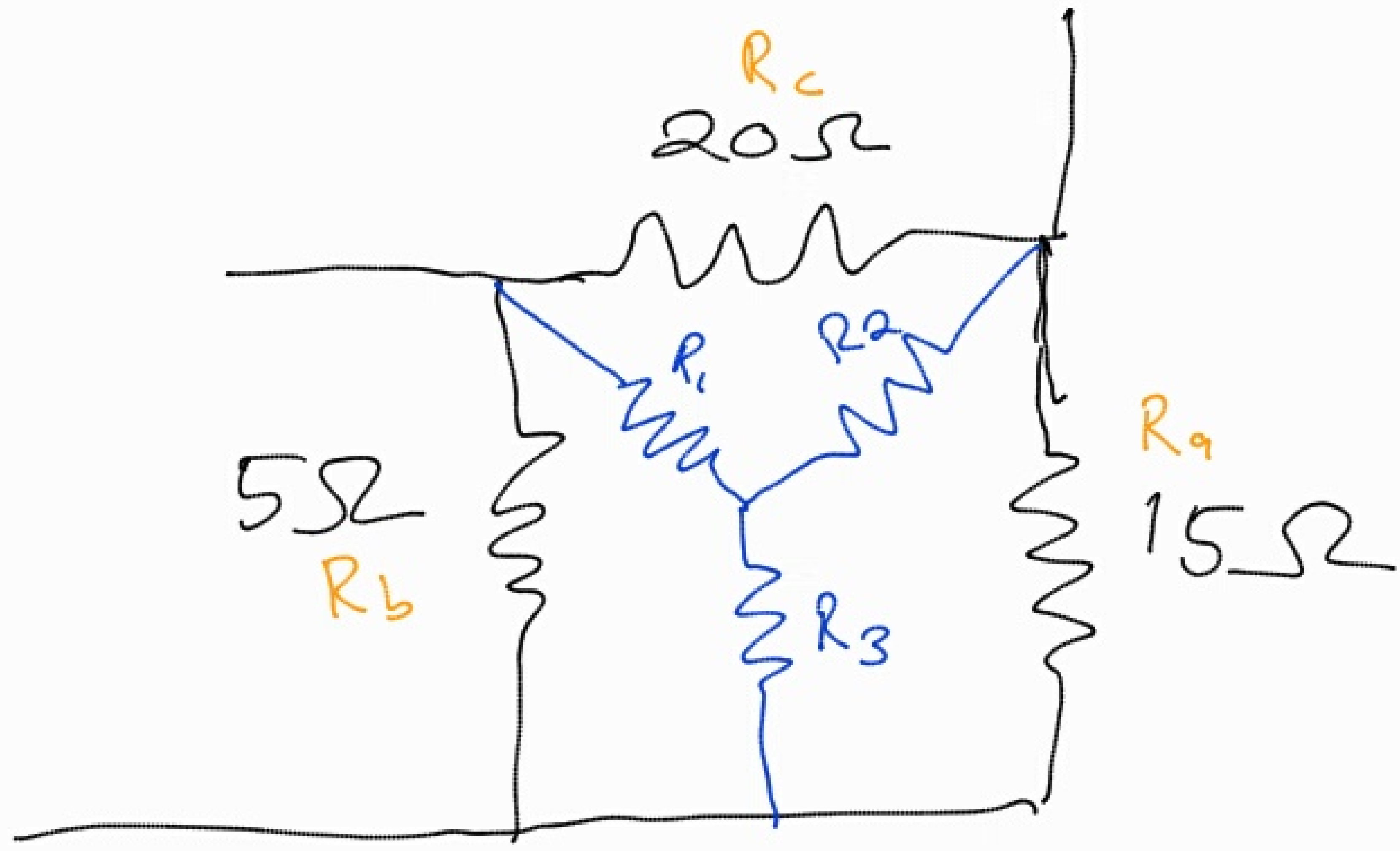
$$= 10{,}4348\,\Omega$$

$$10{,}4348 \parallel 80 = \frac{10{,}4348 \times 80}{10{,}4348 + 80}$$

$$R_{eq} = 9{,}231\,\Omega$$



- There are no parallel or series resistors
- Apply $\Delta \rightarrow Y$ transform



$R_a = 15\,\Omega,\ R_b = 5\,\Omega,\ R_c = 20\,\Omega$

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$= \frac{5 \times 20}{15 + 5 + 20}$$

$$= 2.5\,\Omega$$

$$R_2 = \frac{R_c R_a}{R_a + 5 + 20}$$

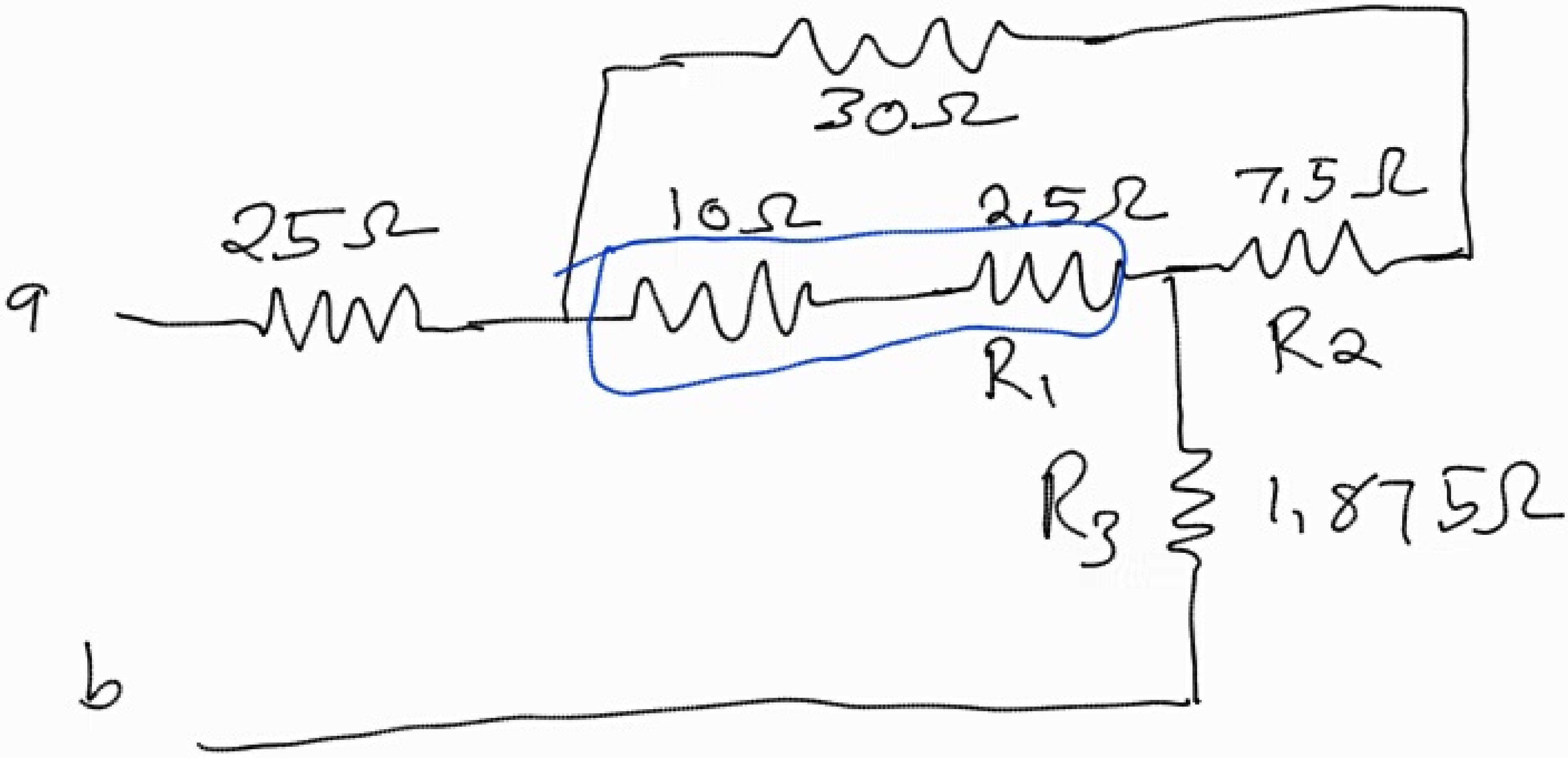
$$= \frac{20 \times 15}{15 + 5 + 20}$$

$$= 7.5\,\Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

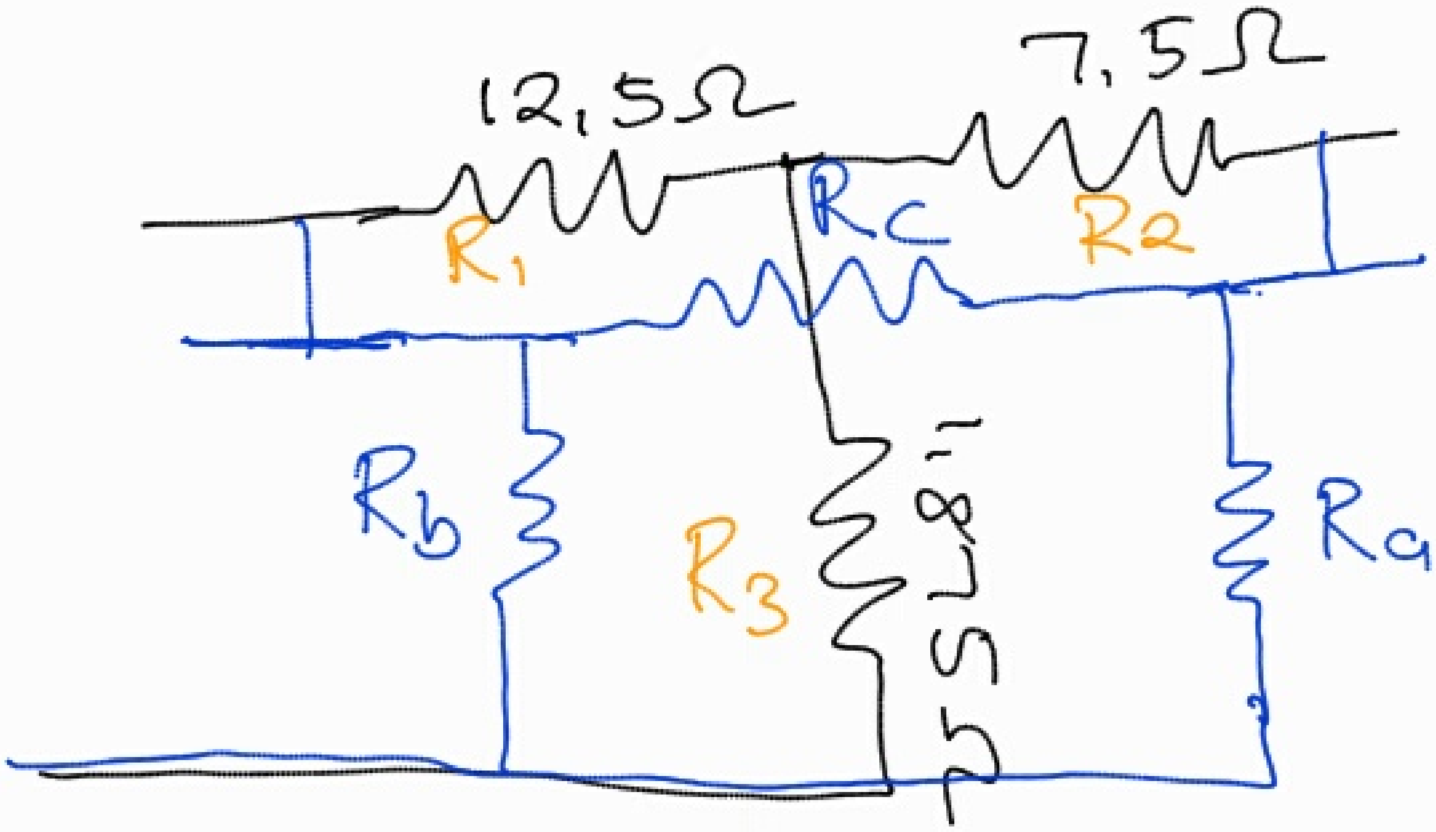
$$= \frac{15 \times 5}{15 + 5 + 20}$$

$$= 1.875\,\Omega$$



$10 + 2,5\,\Omega = 12,5\,\Omega$

Apply $Y \to \Delta$ transform



$$S = R_1R_2 + R_2R_3 + R_3R_1$$

$$= 12{,}5 \times 7{,}5 + 7{,}5 \times 1{,}875 + 1{,}875 \times 12{,}5$$

$$= 131{,}25$$

$$R_a = \frac{S}{R_1}$$

$$= \frac{131{,}25}{12{,}5}$$

$$= 10{,}5\ \Omega$$

$$R_b = \frac{S}{R_2}$$

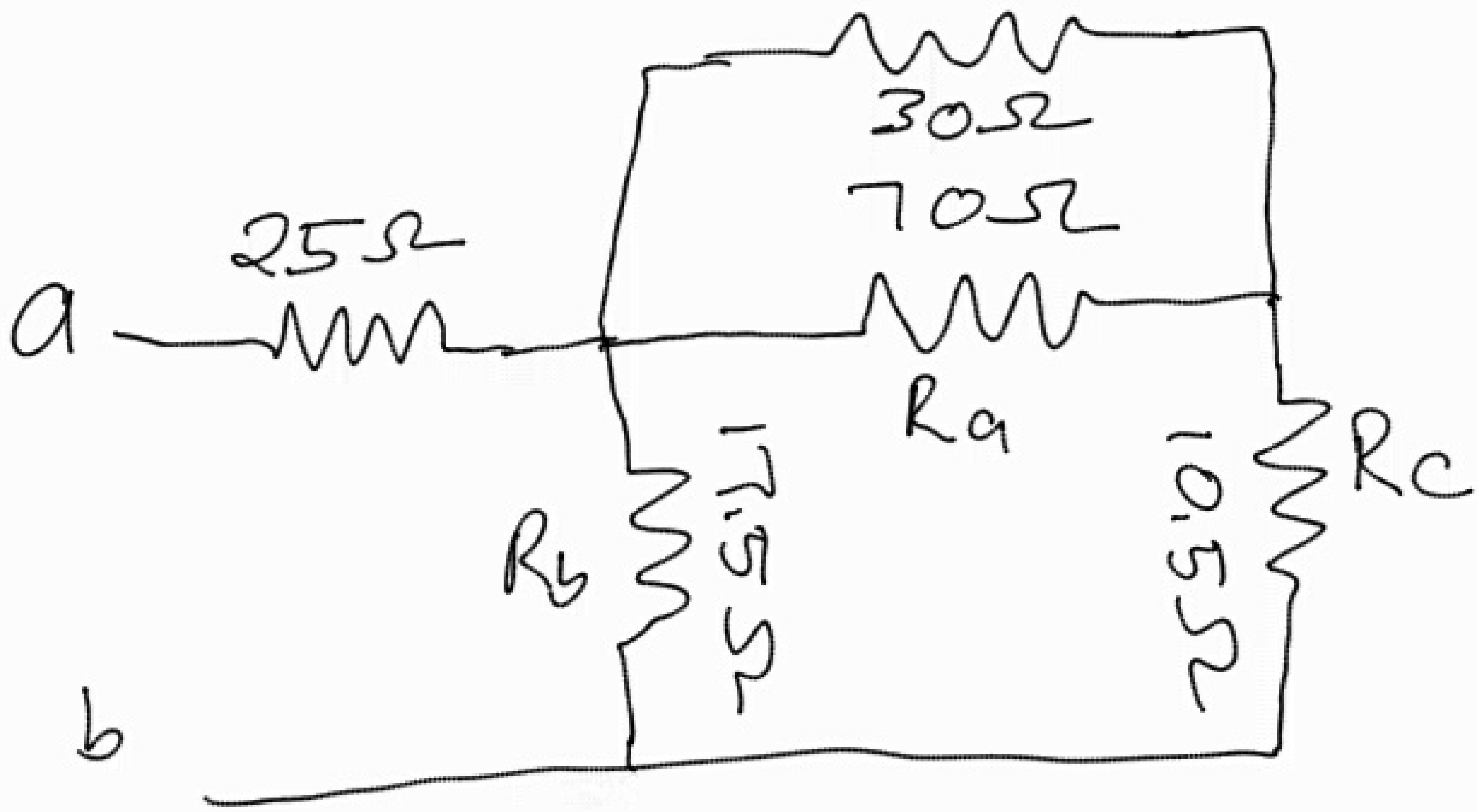
$$= \frac{131{,}25}{7{,}5}$$

$$= 17{,}5\ \Omega$$

$$R_c = \frac{S}{R_3}$$

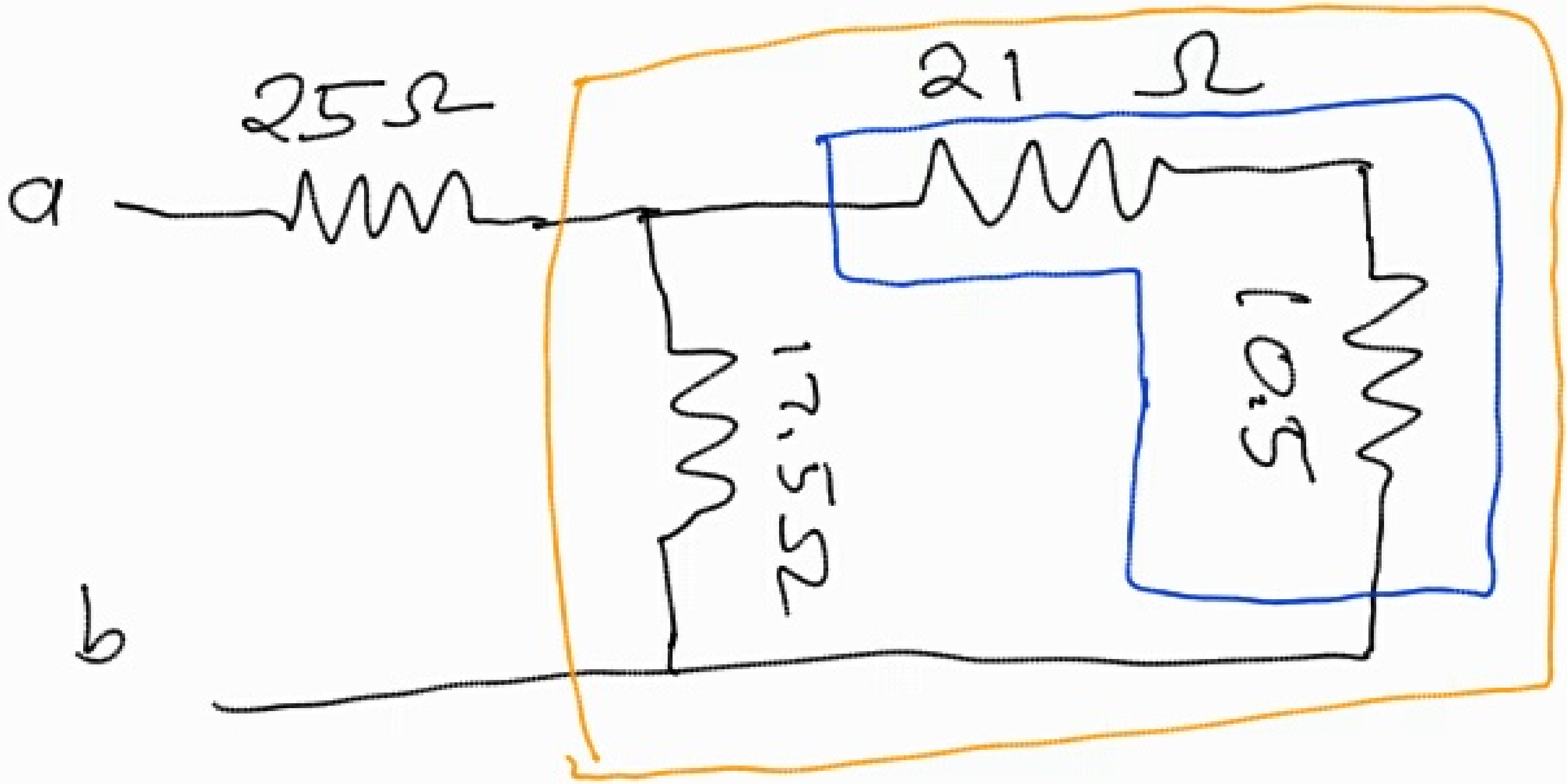
$$= \frac{131{,}25}{1{,}875}$$

$$= 70\ \Omega$$



$$30 \parallel 70 = \frac{30 \times 70}{30 + 70}$$

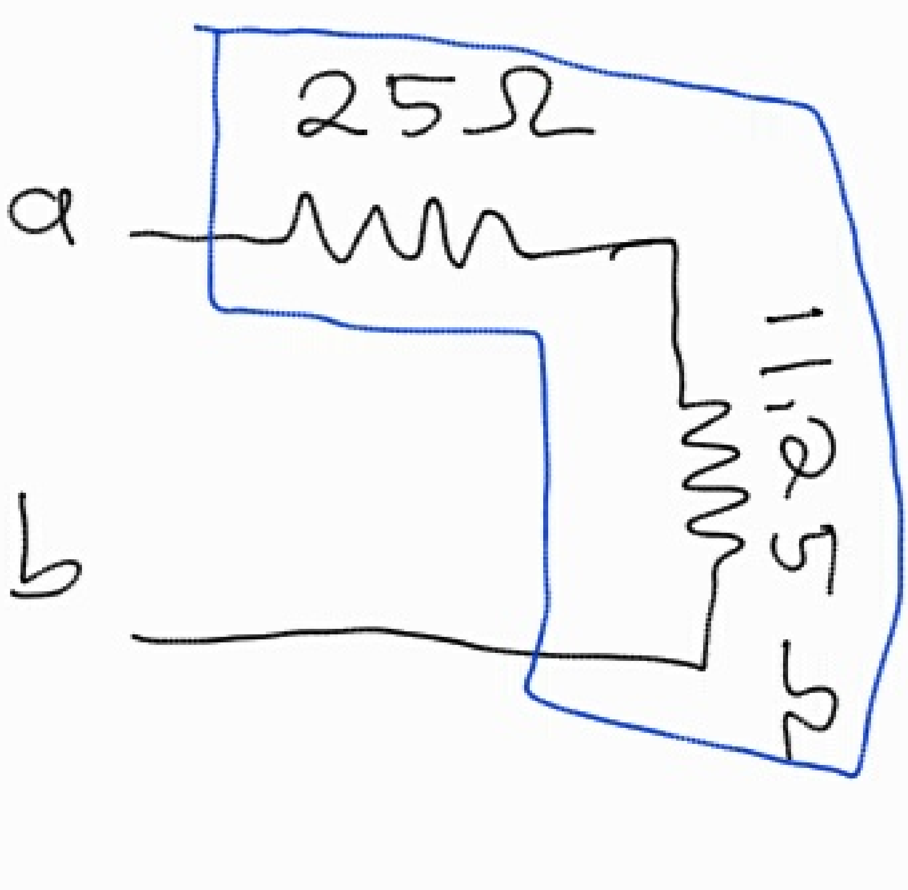
$$= 21 \ \Omega$$



$$21 + 10{,}5 = 31{,}5$$

$$31{,}5 \parallel 17{,}5 = \frac{31{,}5 \times 17{,}5}{31{,}5 + 17{,}5}$$

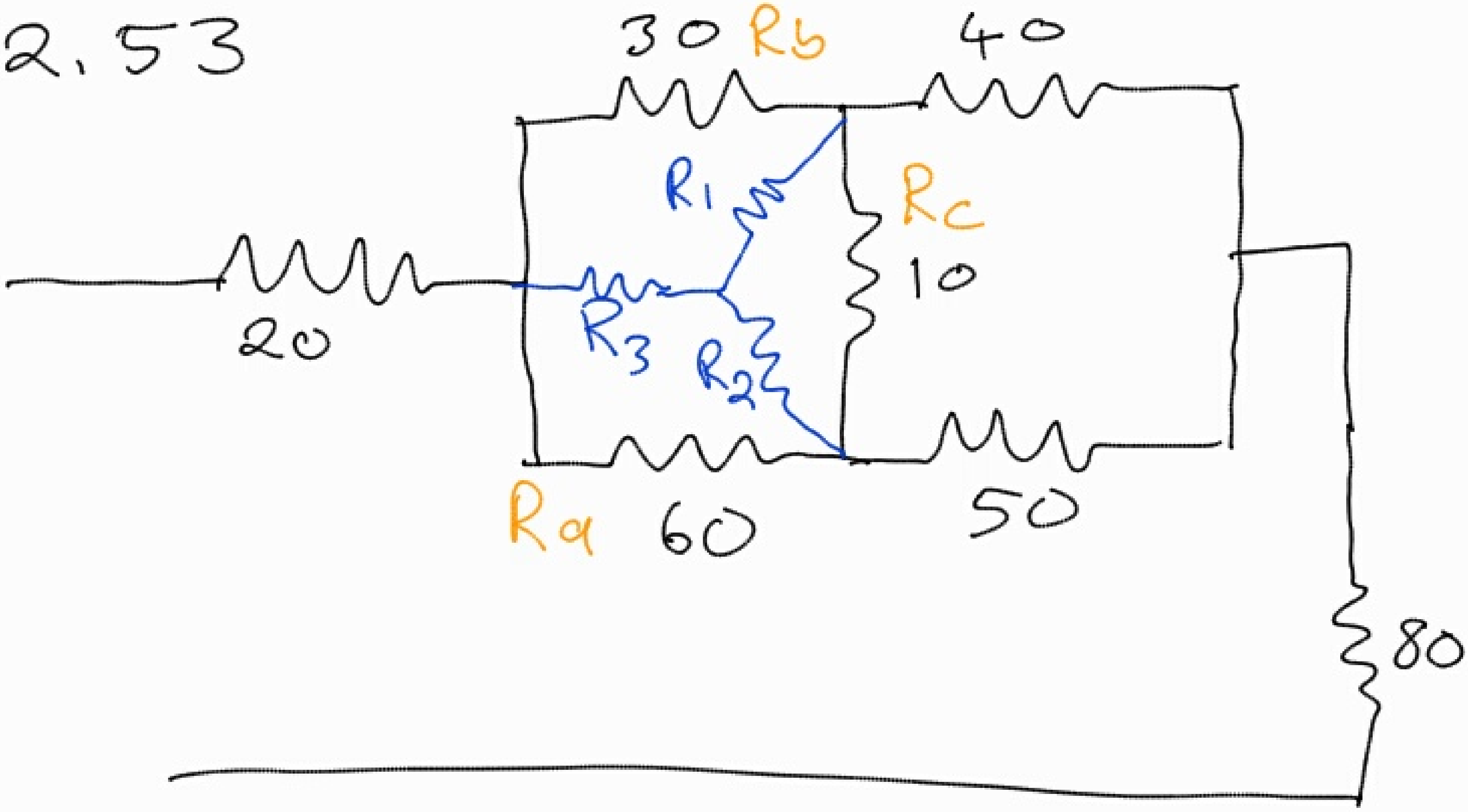
$$= 11{,}25\ \Omega$$



$$R_{eq} = 25 + 11{,}25$$

$$= 36{,}25\ \Omega$$

2.53



$$S = R_a + R_b + R_c$$

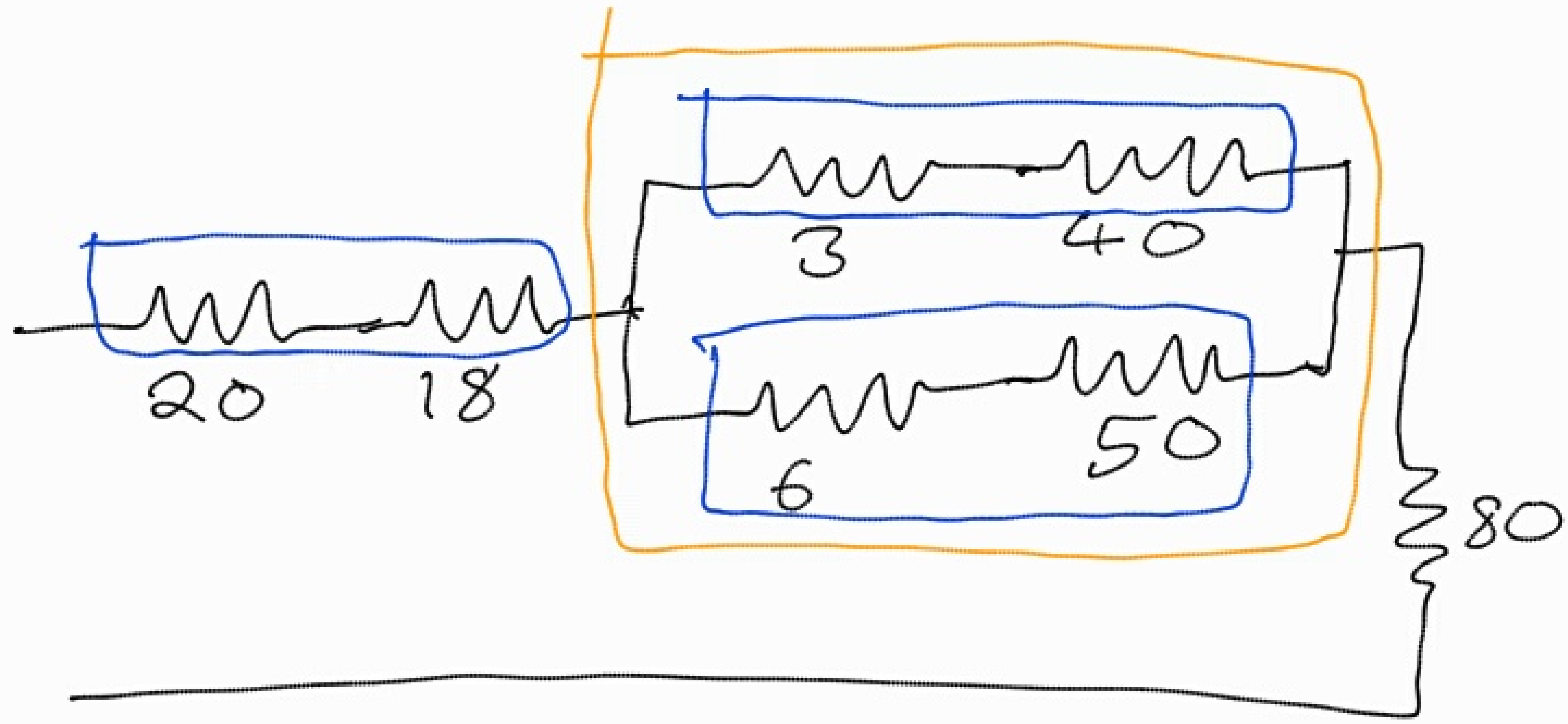
$$= 60 + 30 + 10$$

$$= 100\Omega$$

$$R_1 = \frac{R_b R_c}{S} = \frac{30 \times 10}{100} = 3\Omega$$

$$R_2 = \frac{R_c R_a}{S} = \frac{10 \times 60}{100} = 6\Omega$$

$$R_3 = \frac{R_a R_b}{S} = \frac{60 \times 30}{100} = 18\Omega$$



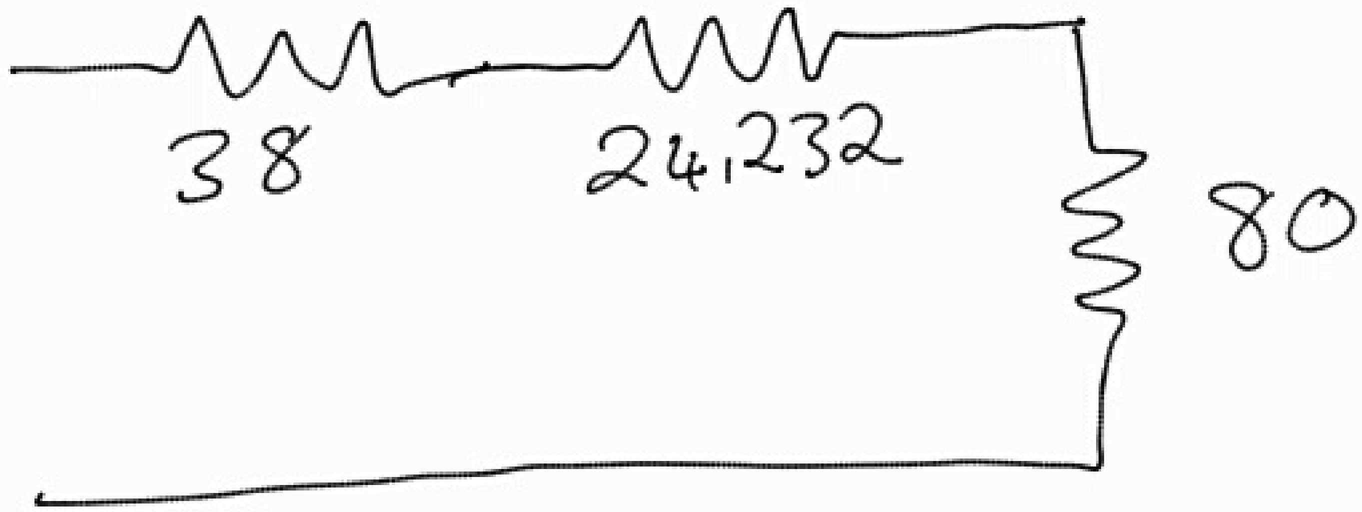
$$20 + 18 = 38\,\Omega$$

$$3 + 40 = 43\,\Omega$$

$$6 + 50 = 56\,\Omega$$

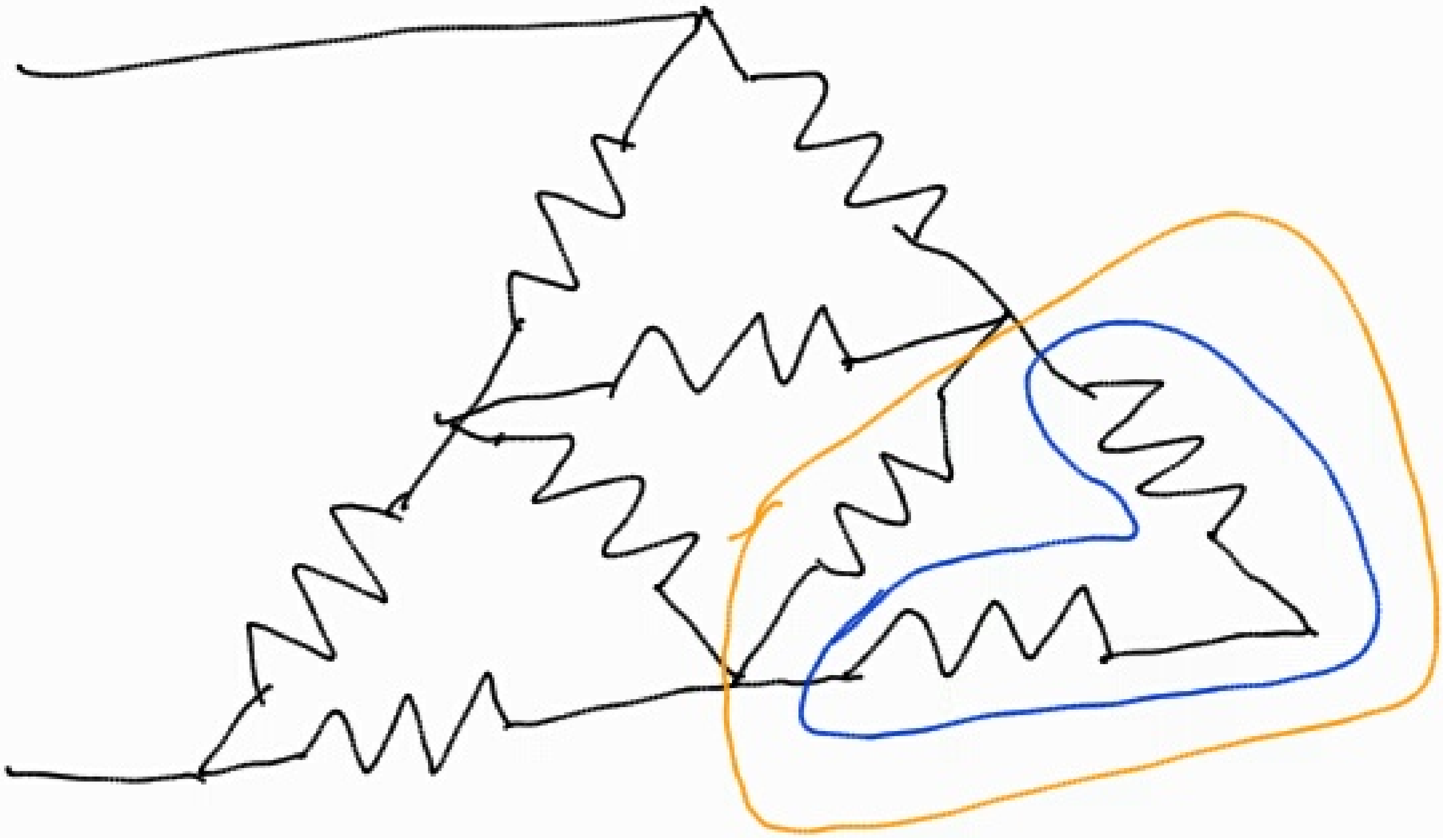
$$43 \parallel 56 = \frac{43 \times 56}{43 + 56}$$

$$= 24{,}232\,\Omega$$



$$R_{eq} = 38 + 24{,}232 + 80$$

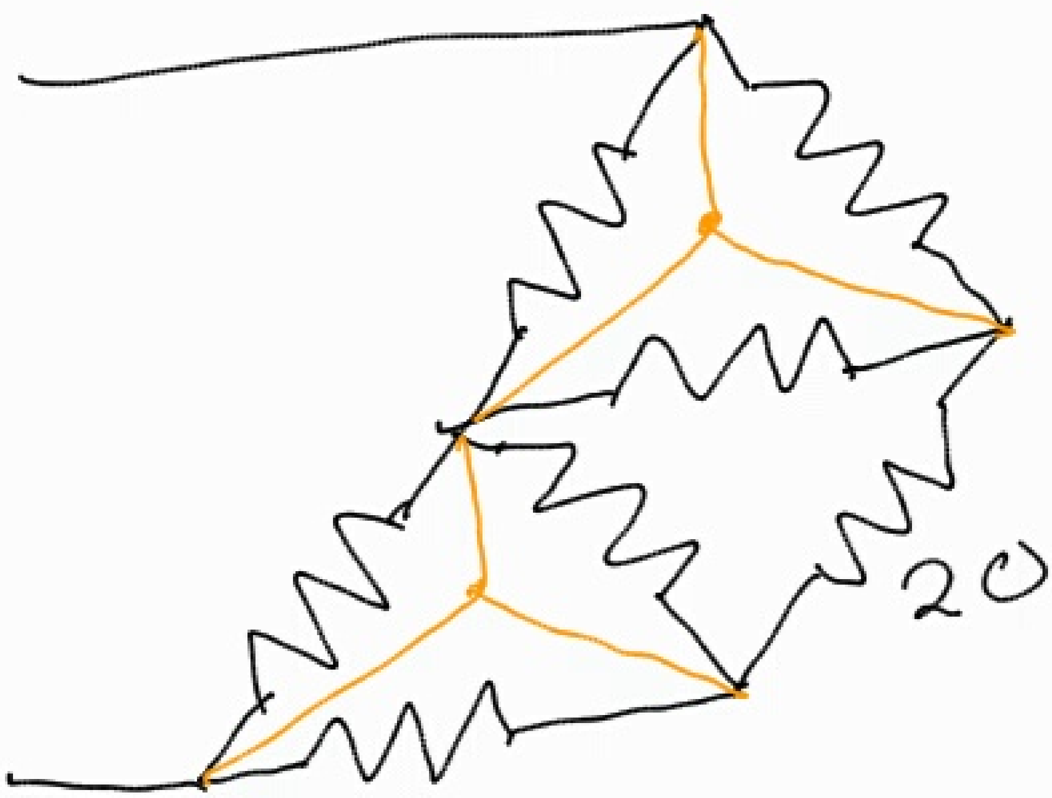
$$= 142{,}323\ \Omega$$



$$30 + 30 = 60\Omega$$

$$30 \parallel 60 = \frac{30 \times 60}{30 + 60}$$

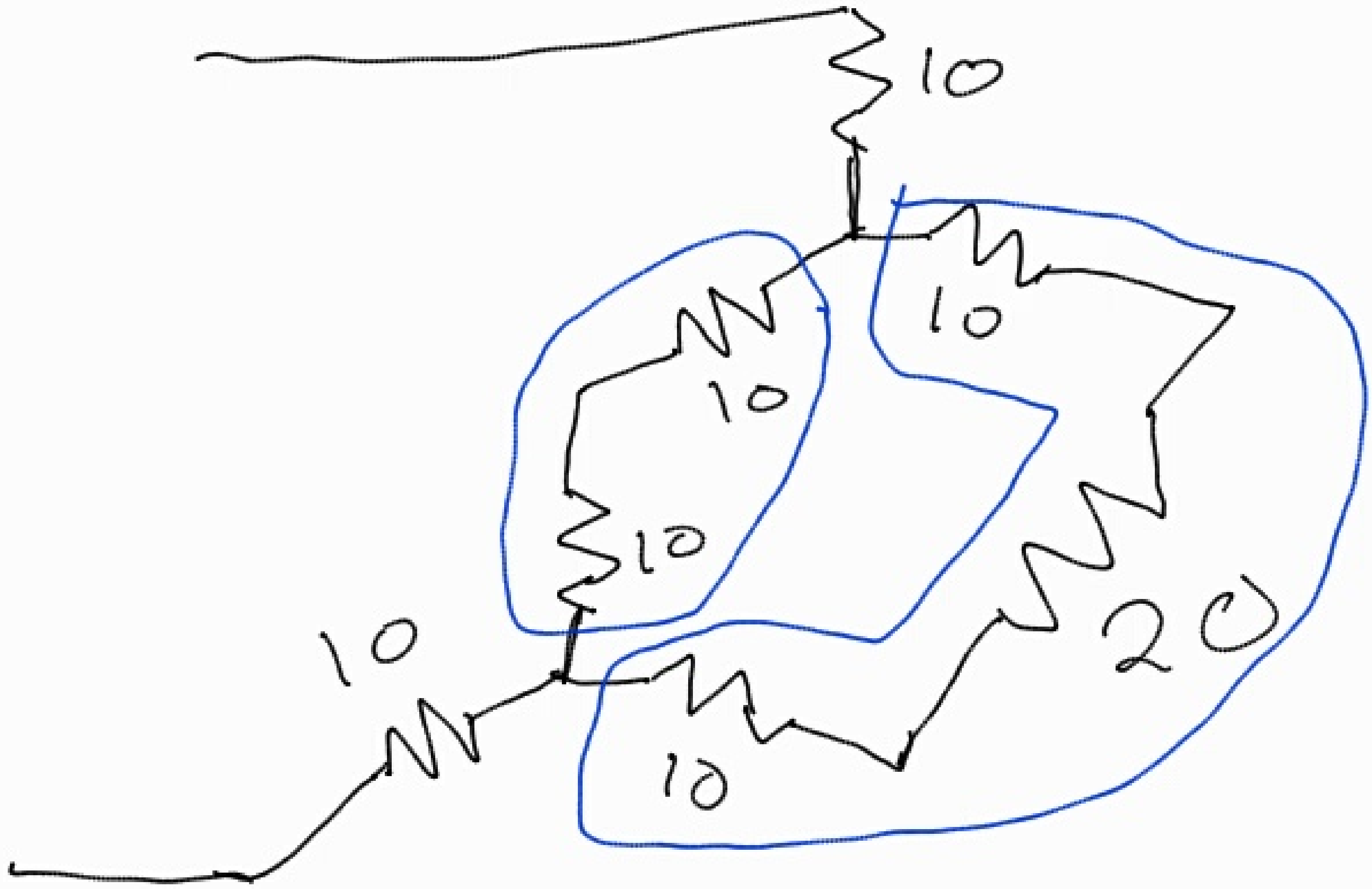
$$= 20\Omega$$



$$R_1 = \frac{R_a R_c}{R_a + R_b + R_c}$$

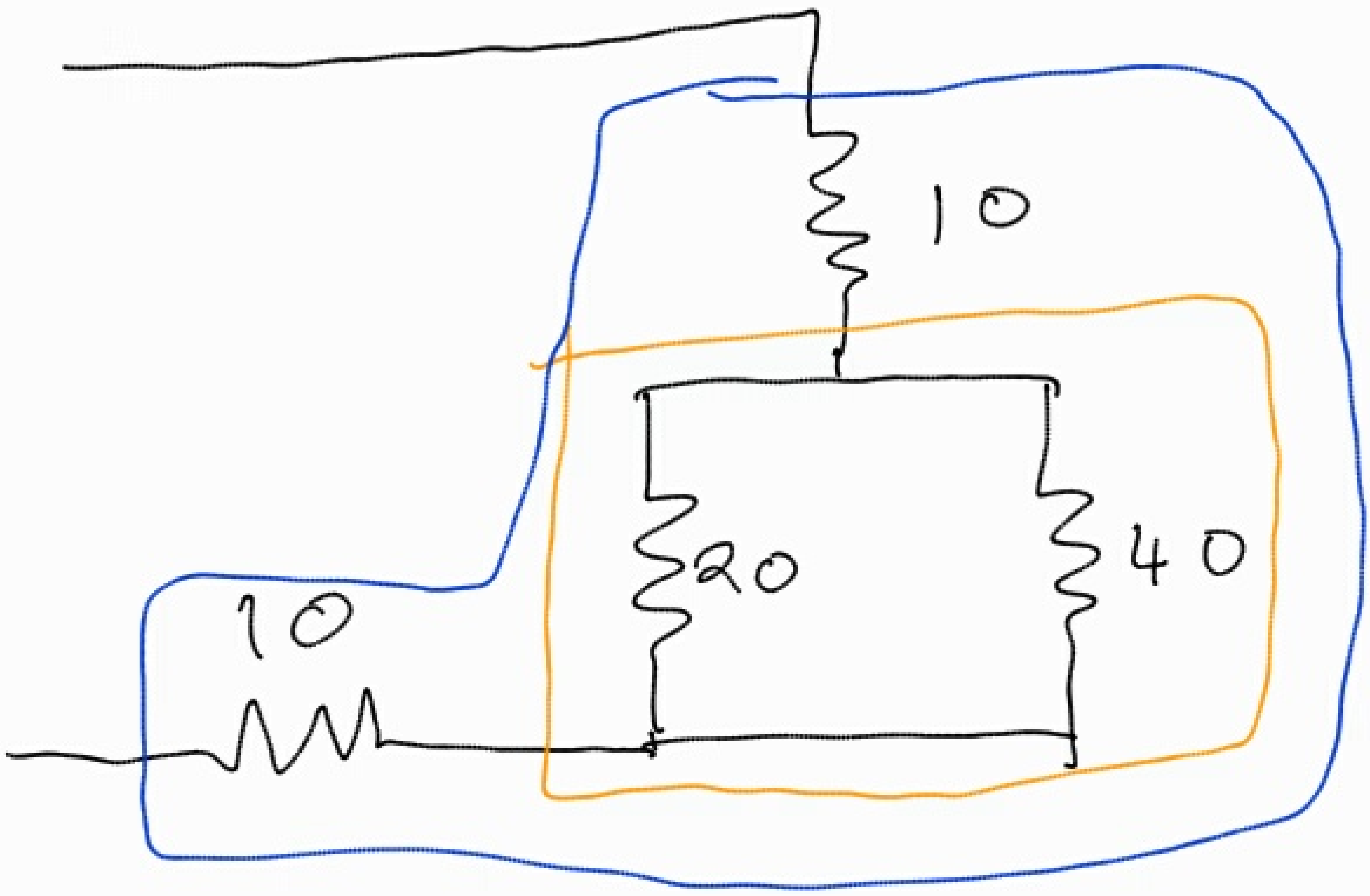
$$= \frac{30 \times 30}{30 + 30 + 30}$$

$$= 10\,\Omega$$



$$10 + 10 = 20\,\Omega$$

$$10 + 20 + 10 = 40\,\Omega$$



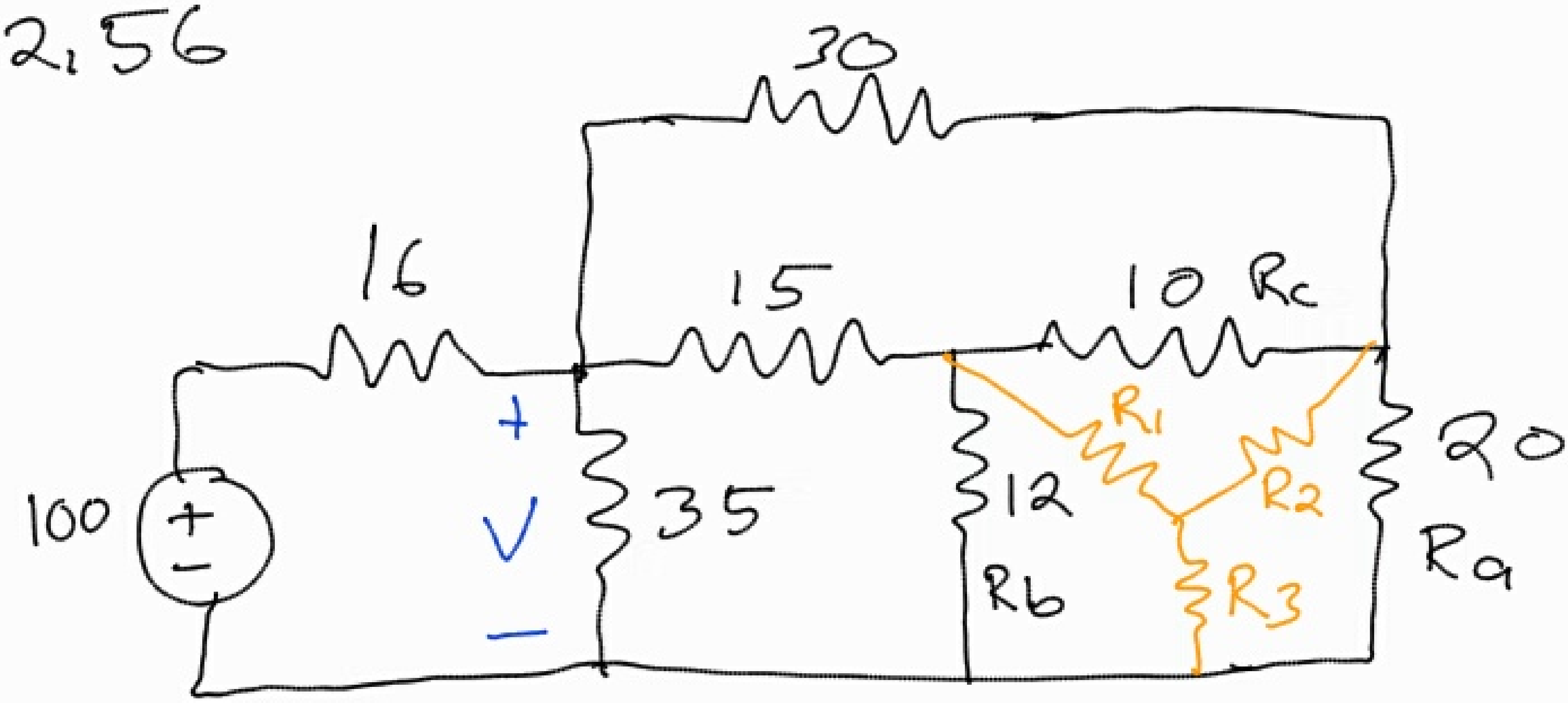
$$20 \| 40 = \frac{20 \times 40}{20 + 40}$$

$$= 13,333$$

$$R_{eq} = 10 + 13,333 + 10$$

$$= 33,333\ \Omega$$

2,56

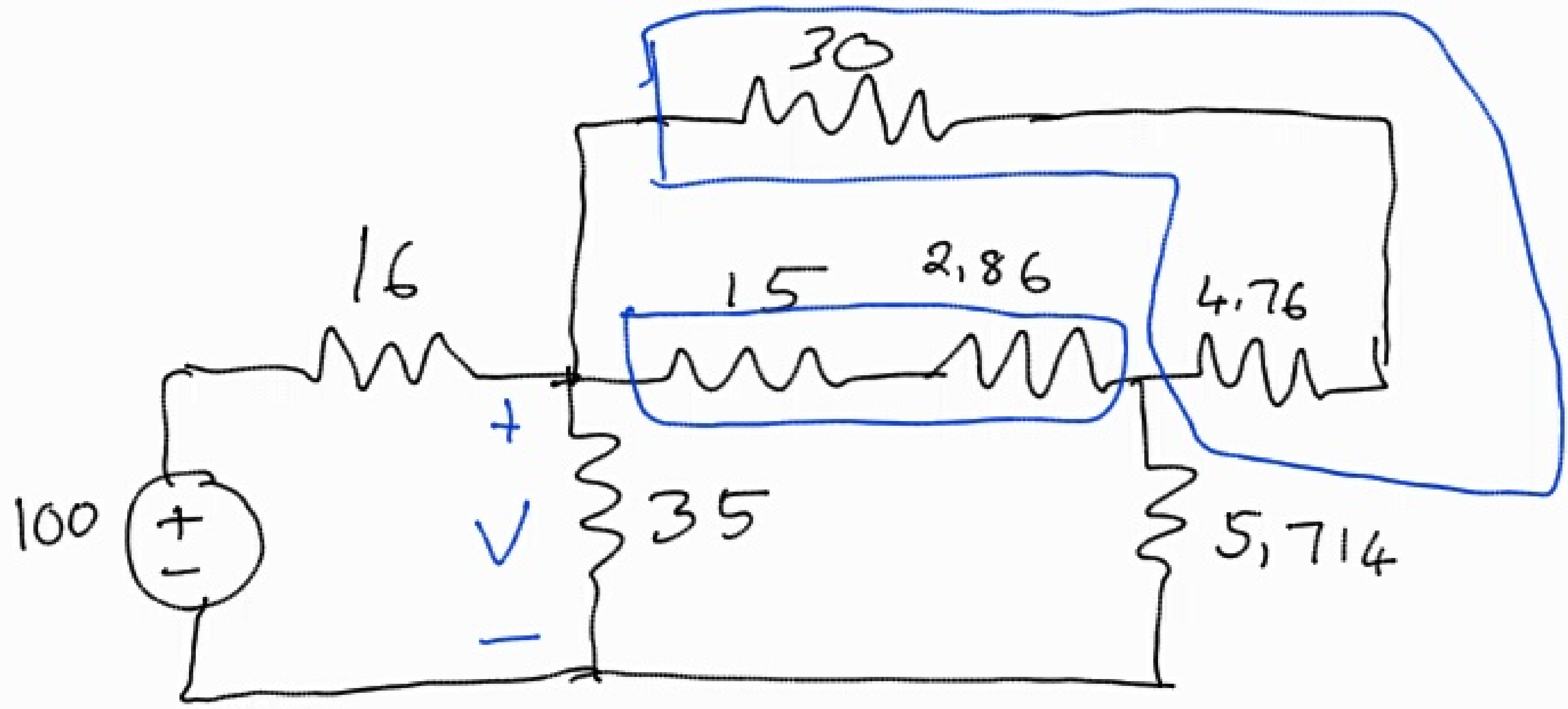


$$S = R_a + R_b + R_c$$
$$= 20 + 12 + 10$$
$$= 42\,\Omega$$

$$R_1 = \frac{R_b R_c}{S} = \frac{12 \times 10}{42} = 2{,}857\,\Omega$$

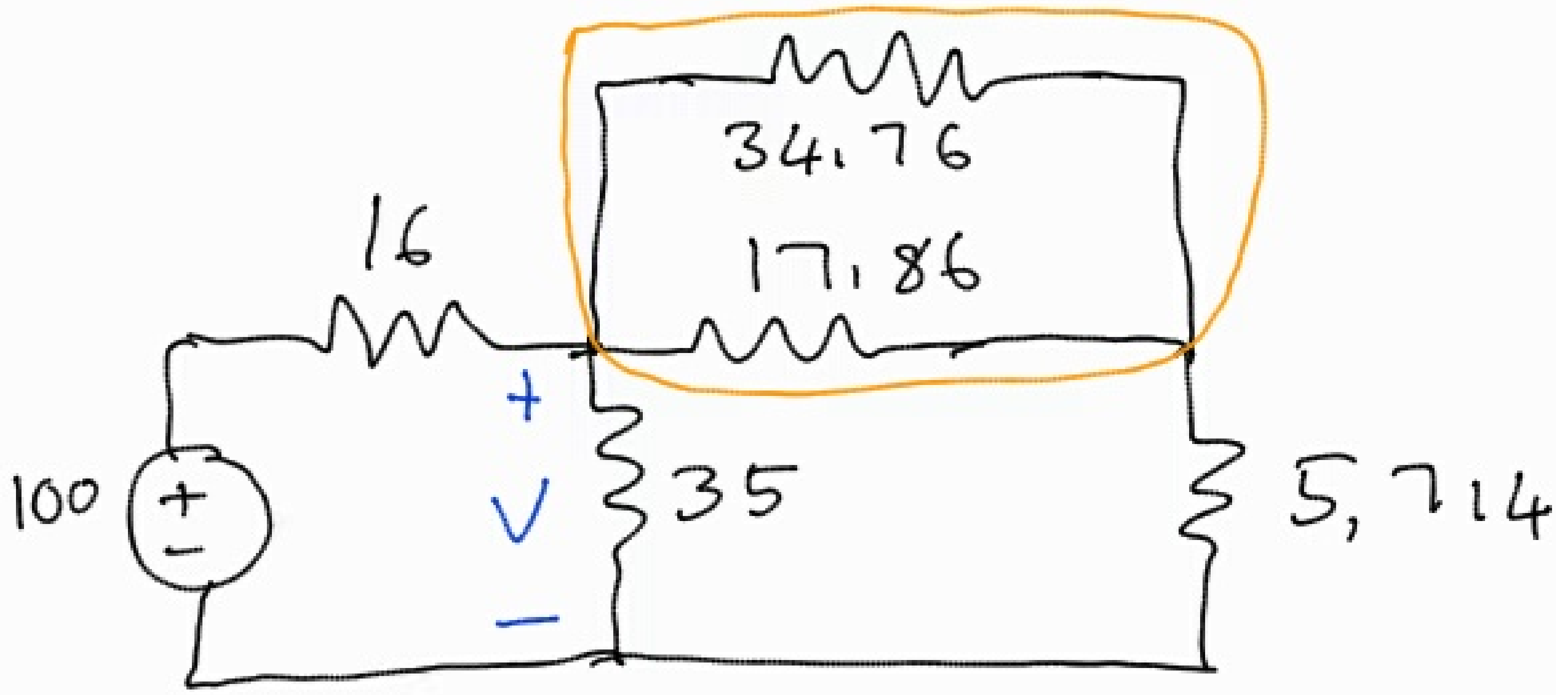
$$R_2 = \frac{R_c R_a}{S} = \frac{10 \times 20}{42} = 4{,}762\,\Omega$$

$$R_3 = \frac{R_a R_b}{S} = \frac{20 \times 12}{42} = 5{,}714\,\Omega$$



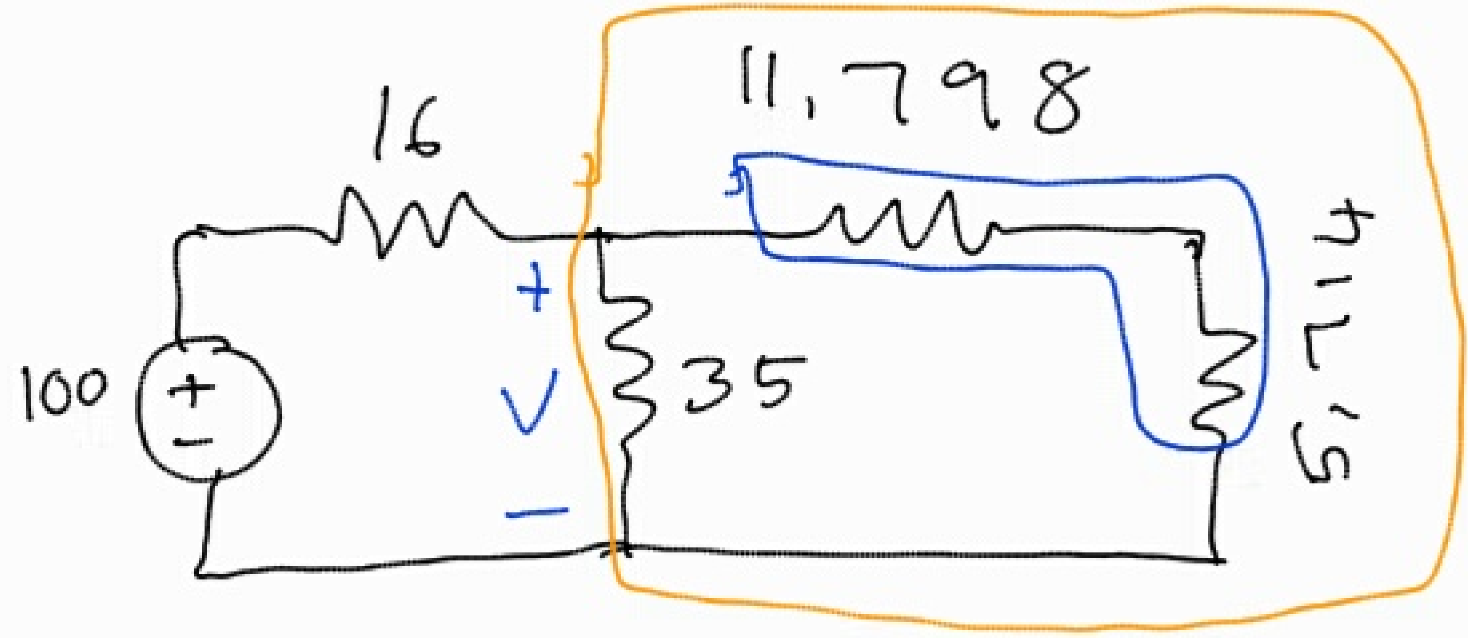
$$15 + 2{,}857 = 17{,}857\,\Omega$$

$$30 + 4{,}762 = 34{,}762$$



$$34{,}76 \parallel 17{,}86 = \frac{34{,}76 \times 17{,}86}{34{,}76 + 17{,}86}$$

$$= 11{,}798\,\Omega$$

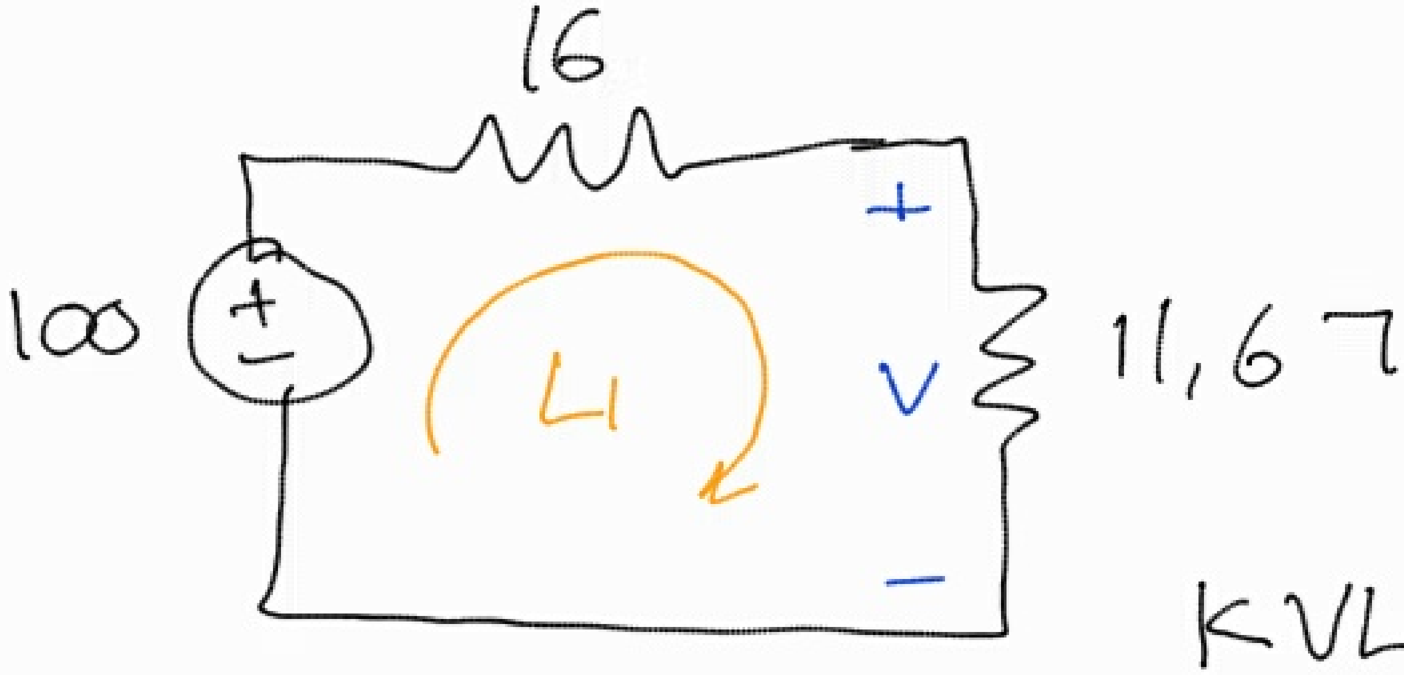


$$11{,}798 + 5{,}714 = 17{,}512$$

$$R_{eq} = 17{,}512 \parallel 35$$

$$= \frac{17{,}512 \times 35}{17{,}512 + 35}$$

$$= 11{,}672$$



KVL ($\Sigma V = 0$)

$$-100 + 16I + 11{,}67I = 0$$

$$27{,}67I = 100$$

$$I = \frac{100}{27{,}67}$$

$$I = 3{,}614 \text{ A}$$

$V = IR$ (Ohm's law)

$$V = 3{,}614 \times 11{,}67$$

$$= 42{,}18 \text{ V}$$